

Keysight Models 66312A, 66332A
Dynamic Measurement DC Sources
Keysight Models 6631B, 6632B,
6633B, 6634B, 6611C, 6612C,
6613C, and 6614C
System DC Power Supplies

Programming Guide

NOTICE: This document contains references to Agilent Technologies. Agilent's former Test and Measurement business has become Keysight Technologies. For more information, go to **www.keysight.com**.



Safety Guidelines

The beginning of the Operating Guide has a Safety Summary page. Be sure you are familiar with the information on this page before programming the dc source for operation from a controller.

Printing History

The edition and current revision of this manual are indicated below. Reprints of this guide containing minor corrections and updates may have the same printing date. Revised editions are identified by a new printing date. A revised edition incorporates all new or corrected material since the previous printing date. Changes to the manual occurring between revisions are covered by change sheets shipped with the guide.

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General Information

About this Guide

This guide provides remote programming information for the following series of GPIB programmable dc power supplies:

- Keysight 66312A
- Keysight 66332A
- Keysight 6631B/6632B/6633B/6634B
- Keysight 6611C/6612C/6613C/6614C

You will find the following information in the rest of this guide:

Chapter 1	Introduction to this guide.
Chapter 2	Introduction to SCPI messages structure, syntax, and data formats. Examples of SCPI programs
Chapter 3	Introduction to Programming the dc source with SCPI commands.
Chapter 4	Dictionary of SCPI commands.
Appendix A	SCPI conformance information.
Appendix B	Use of the alternate Comptibility programming language.
Appendix C	Error messages

Documentation Summary

The following documents that are related to this Programming Guide have additional helpful information for using the dc source.

- ◆ User's Guide for Keysight 66312A and Keysight 6611C/6612C/6613C/3314C. Includes specifications and supplemental characteristics, how to use the front panel, how to connect to the instrument, and calibration procedures.
- ◆ User's Guide for Keysight 66332A and Keysight 6631B/6632B/6633B/6634B. Includes specifications and supplemental characteristics, how to use the front panel, how to connect to the instrument, and calibration procedures.

External References

GPIB References

The most important GPIB documents are your controller programming manuals - BASIC, GPIB Command Library for MS DOS, etc. Refer to these for all non-SCPI commands (for example: Local Lockout).

The following are two formal documents concerning the GPIB interface:

- ◆ *ANSI/IEEE Std. 488.1-1987 IEEE Standard Digital Interface for Programmable Instrumentation.* Defines the technical details of the GPIB interface. While much of the information is beyond the need of most programmers, it can serve to clarify terms used in this guide and in related documents.
- ◆ *ANSI/IEEE Std. 488.2-1987 IEEE Standard Codes, Formats, Protocols, and Common Commands.* Recommended as a reference only if you intend to do fairly sophisticated programming. Helpful for finding precise definitions of certain types of SCPI message formats, data types, or common commands.

The above two documents are available from the IEEE (Institute of Electrical and Electronics Engineers), 345 East 47th Street, New York, NY 10017, USA. The WEB address is www.ieee.org.

SCPI References

The following documents will assist you with programming in SCPI:

- ◆ *Standard Commands for Programmable Instruments Volume 1, Syntax and Style*
- ◆ *Standard Commands for Programmable Instruments Volume 2, Command References*
- ◆ *Standard Commands for Programmable Instruments Volume 3, Data Interchange Format*
- ◆ *Standard Commands for Programmable Instruments Volume 4, Instrument Classes*

To obtain a copy of the above documents, contact: Fred Bode, Executive Director, SCPI Consortium, 8380 Hercules Drive, Suite P3, Ls Mesa, CA 91942, USA

Introduction to Programming

VXIplug&play Power Products Instrument Drivers

VXIplug&play instrument drivers for Microsoft Windows 95 and Windows NT are now available on the Web at <http://www.keysight.com/find/drivers>. These instrument drivers provide a high-level programming interface to your Keysight Technologies instrument. VXIplug&play instrument drivers are an alternative to programming your instrument with SCPI command strings. Because the instrument driver's function calls work together on top of the VISA I/O library, a single instrument driver can be used with multiple application environments.

Supported Applications

- Keysight VEE
- Microsoft Visual BASIC
- Microsoft Visual C/C++
- Borland C/C++
- National Instruments LabVIEW
- National Instruments LabWindows/CVI

System Requirements

The VXIplug&play Power Products instrument driver complies with the following:

- Microsoft Windows 95
- Microsoft Windows NT 4.0
- HP VISA revision F.01.02
- National Instruments VISA 1.1

Downloading and Installing the Driver

NOTE: Before installing the VXIplug&play instrument driver, make sure that you have one of the supported applications installed and running on your computer.

1. Access Keysight Technologies' Web site at <http://www.keysight.com/find/drivers>.
2. Select the instrument for which you need the driver.
3. Click on the driver, either Windows 95 or Windows NT, and download the executable file to your pc.
4. Locate the file that you downloaded from the Web. From the **Start** menu select **Run** <path>:\agxxxx.exe - where <path> is the directory path where the file is located, and agxxxx is the instrument driver that you downloaded .
5. Follow the directions on the screen to install the software. The default installation selections will work in most cases. The readme.txt file contains product updates or corrections that are not documented in the on-line help. If you decide to install this file, use any text editor to open and read it.

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6. To use the *VXIplug&play* instrument driver, follow the directions in the *VXIplug&play* online help under "Introduction to Programming".

Accessing Online Help

A comprehensive online programming reference is provided with the driver. It describes how to get started using the instrument driver with Keysight VEE, LabVIEW, and LabWindows. It includes complete descriptions of all function calls as well as example programs in C/C++ and Visual BASIC.

- To access the online help when you have chosen the default Vxipnp start folder, click on the Start button and select Programs | Vxipnp | Agxxxx Help (32-bit).
- where agxxxx is the instrument driver.

GPIO Capabilities of the DC Source

All dc source functions except for setting the GPIO address are programmable over the GPIO. The IEEE 488.2 capabilities of the dc source are listed in the Specifications Table of the User's Guide.

GPIO Address

The dc source operates from an GPIO address that is set from the front panel. To set the GPIO address, press the **Address** key on the front panel and enter the address using the Entry keys. The GPIO address is stored in non-volatile memory.

RS-232 Capabilities of the DC Source

The dc source provides an RS-232 programming interface, which is activated by commands located under the front panel **Address** key. All SCPI and COMPatibility commands are available through RS-232 programming. When the RS-232 interface is selected, the GPIO interface is disabled.

The EIA RS-232 Standard defines the interconnections between Data Terminal Equipment (DTE) and Data Communications Equipment (DCE). The dc source is designed to be a DTE. It can be connected to another DTE such as a PC COM port through a null modem cable.

NOTE: The RS-232 settings in your program must match the settings specified in the front panel Address menu. Press the front panel Address key if you need to change the settings.

RS-232 Data Format

The RS-232 data is a 10-bit word with one start bit and one stop bit. The number of start and stop bits is not programmable. However, the following parity options are selectable using the front panel Address key:

EVEN	Seven data bits with even parity
ODD	Seven data bits with odd parity
MARK	Seven data bits with mark parity (parity is always true)
SPACE	Seven data bits with space parity (parity is always false)
NONE	Eight data bits without parity

Parity options are stored in non-volatile memory.

Baud Rate

The front panel Address key lets you select one of the following baud rates, which is stored in non-volatile memory:

300 600 1200 2400 4800 9600

RS-232 Flow Control

The RS-232 interface supports several flow control options that are selected using the front panel Address key. For each case, the dc source will send a maximum of five characters after holdoff is asserted by the controller. The dc source is capable of receiving as many as fifteen additional characters after it asserts holdoff.

XON-XOFF	A software handshake that uses the ASCII control code DC3 (decimal code 19) to assert hold-off, and control code DC1 (decimal code 17) to release hold-off.
RTS-CTS	The dc source asserts its Request to Send (RTS) line to signal hold-off when its input buffer is almost full, and it interprets its Clear to Send (CTS) line as a hold-off signal from the controller.
DTR-DSR	The dc source asserts its Data Terminal Ready (DTR) line to signal hold-off when its input buffer is almost full, and it interprets its Data Set Ready (DSR) line as a hold-off signal from the controller.
NONE	There is no flow control.

Flow control options are stored in non-volatile memory.

RS-232 Programming Example

The following program illustrates how to program the power supply using RS-232 to set the output voltage and current and to readback the model number and output voltage. The program was written to run on any controller using Microsoft QBasic.

NOTE: The power supply must be configured for RS232 and the same baud rate and parity as the controller.

```

\ Program to write and read via RS232.
\ Configure the power supply for 9600 baud, even parity and RS232
\ Configure serial port for:"
\   9600 baud
\   7 bit data
\   2 stop bits
\   Ignore request to send
\   Ignore carrier detect
\   Even parity           \ Needed with Vectra basic, ignored with QBasic
\   Send line feed
\   Reserve 1000 character buffer for serial I/O
\
DECLARE FUNCTION gets$ ()
CLS                               \ Clears screen
LOCATE 1, 1                       \ Position cursor at top left
\ Configure Com Port
OPEN "com1:9600,e,7,2,rs,cd,pe,lf" FOR RANDOM AS #1 LEN = 1000
PRINT #1, "OUTPUT ON"           \ Turn on output then set voltage and current
PRINT #1, "VOLT 6"               \ Set voltage to 6 volts
PRINT #1, "CURR .5"              \ Set current to 0.5 amps
PRINT #1, "*IDN?"                \ Query the power supply identification string
PRINT gets$                     \ Go to gets$ Function and print data returned
PRINT #1, MEAS"VOLT?"; volt      \ Query the power supply voltage
VOLT = VAL (gets$)               \ Convert gets$ string to a value
PRINT gets$                     \ Print the value of the voltage
END                             \ End of main program

```

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```
FUNCTION gets$                                ' Get a new line feed terminated string from device #1
C$ = ""                                       ' Set C$ to null
WHILE c$ <> CHR$ (10)                         ' Set loop to stop at Line Feed
    C$ = INPUT$ (1, #1)                     ' Read 1 bit into file #1
    Resp$ = resp$ + c$                     ' Concatenate bit with previous bits
WEND                                         ' End of WHILE loop
gets$ = resp$                               ' Assign response to gets$
END FUNCTION
```

RS-232 Troubleshooting

If you are having trouble communicating over the RS-232 interface, check the following:

- ◆ The computer and the dc source must be configured for the same baud rate, parity, number of data bits, and flow control options. Note that the dc source is configured for 1 start bit and 1 stop bit (these values are fixed).
- ◆ The correct interface cables or adaptors must be used, as described under RS-232 Connector. Note that even if the cable has the proper connectors for your system, the internal wiring may be incorrect.
- ◆ The interface cable must be connected to the correct serial port on your computer (COM1, COM2, etc.).

Introduction to SCPI

SCPI (Standard Commands for Programmable Instruments) is a programming language for controlling instrument functions over the GPIB. SCPI is layered on top of the hardware-portion of IEEE 488.2. The same SCPI commands and parameters control the same functions in different classes of instruments. For example, you would use the same DISPlay command to control the dc source display and the display of a SCPI-compatible multimeter.

Conventions Used in This Guide

Angle brackets	< >	Items within angle brackets are parameter abbreviations. For example, <NR1> indicates a specific form of numerical data.
Vertical bar		Vertical bars separate alternative parameters. For example, NORM TEXT indicates that either "TEXT" or "NORM" can be used as a parameter.
Square Brackets	[]	Items within square brackets are optional. The representation [SOURce:]. VOLTage means that SOURce: may be omitted.
Braces	{ }	Braces indicate parameters that may be repeated zero or more times. It is used especially for showing arrays. The notation <A>{} shows that parameter "A" must be entered, while parameter "B" may be omitted or may be entered one or more times.
Boldface font		Boldface font is used to emphasize syntax in command definitions. TRIGger:COUNT:CURRent <NRf> shows command definition.
Computer font		Computer font is used to show program lines in text. TRIGger:COUNT:CURRent 10 shows a program line.

Types of SCPI Commands

SCPI has two types of commands, common and subsystem.

- ◆ Common commands generally are not related to specific operation but to controlling overall dc source functions, such as reset, status, and synchronization. All common commands consist of a three-letter mnemonic preceded by an asterisk: *RST *IDN? *SRE 8
- ◆ Subsystem commands perform specific dc source functions. They are organized into an inverted tree structure with the "root" at the top. The following figure shows a portion of a subsystem command tree, from which you access the commands located along the various paths. You can see the complete tree in Table A-1

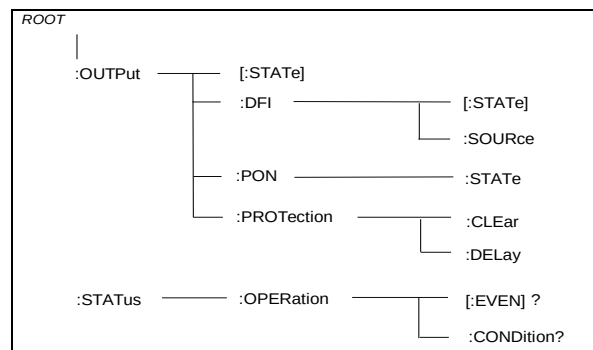


Figure 2-1. Partial Command Tree

Multiple Commands in a Message

Multiple SCPI commands can be combined and sent as a single message with one message terminator. There are two important considerations when sending several commands within a single message:

- ◆ Use a semicolon to separate commands within a message.
- ◆ There is an implied header path that affects how commands are interpreted by the dc source.

The header path can be thought of as a string that gets inserted **before** each command within a message. For the first command in a message, the header path is a null string. For each subsequent command the header path is defined as the characters that make up the headers of the previous command in the message up to and including the last colon separator. An example of a message with two commands is:

```
CURR:LEV 3;PROT:STAT OFF
```

which shows the use of the semicolon separating the two commands, and also illustrates the header path concept. Note that with the second command, the leading header "CURR" was omitted because after the "CURR:LEV 3" command, the header path was became defined as "CURR" and thus the instrument interpreted the second command as:

```
CURR:PROT:STAT OFF
```

In fact, it would have been syntactically incorrect to include the "CURR" explicitly in the second command, since the result after combining it with the header path would be:

```
CURR:CURR:PROT:STAT OFF
```

which is incorrect.

Moving Among Subsystems

In order to combine commands from different subsystems, you need to be able to reset the header path to a null string within a message. You do this by beginning the command with a colon (:), which discards any previous header path. For example, you could clear the output protection and check the status of the Operation Condition register in one message by using a root specifier as follows:

```
OUTPut:PROTection:CLEAr;:STATus:OPERation:CONDition?
```

The following message shows how to combine commands from different subsystems as well as within the same subsystem:

```
VOLTage:LEVel 20;PROTection 28; :CURRent:LEVel 3;PROTection:STATe ON
```

Note the use of the optional header LEVel to maintain the correct path within the voltage and current subsystems, and the use of the root specifier to move between subsystems.

Including Common Commands

You can combine common commands with system commands in the same message. Treat the common command as a message unit by separating it with a semicolon (the message unit separator). Common commands *do not affect the header path*; you may insert them anywhere in the message.

```
VOLTage:TRIGgered 17.5;:INITialize;*TRG  
OUTPut OFF;*RCL 2;OUTPut ON
```

Using Queries

Observe the following precautions with queries:

- ◆ Set up the proper number of variables for the returned data.
- ◆ Read back all the results of a query before sending another command to the dc source. Otherwise a *Query Interrupted* error will occur and the unreturned data will be lost.

Types of SCPI Messages

There are two types of SCPI messages, program and response.

- ◆ A *program message* consists of one or more properly formatted SCPI commands sent from the controller to the dc source. The message, which may be sent at any time, requests the dc source to perform some action.
- ◆ A *response message* consists of data in a specific SCPI format sent from the dc source to the controller. The dc source sends the message only when commanded by a program message called a "query."

The following figure illustrates SCPI message structure:

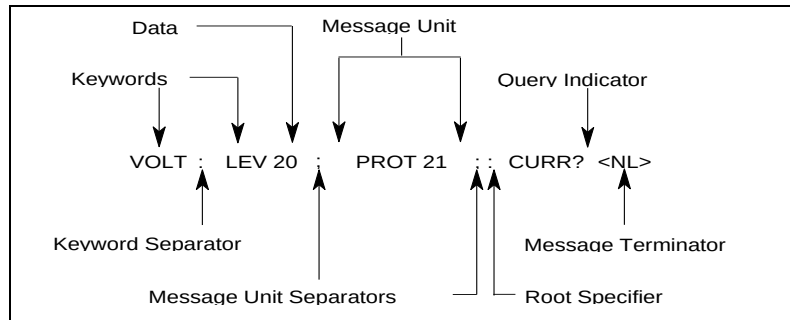


Figure 2-2. Command Message Structure

The Message Unit

The simplest SCPI command is a single message unit consisting of a command header (or keyword) followed by a message terminator. The message unit may include a parameter after the header. The parameter can be numeric or a string.

```
ABORt<NL>
VOLTage 20<NL>
```

Headers

Headers, also referred to as keywords, are instructions recognized by the dc source. Headers may be either in the long form or the short form. In the long form, the header is completely spelled out, such as VOLTAGE, STATUS, and DELAY. In the short form, the header has only the first three or four letters, such as VOLT, STAT, and DEL.

Query Indicator

Following a header with a question mark turns it into a query (VOLTage?, VOLTage:PROTection?). If a query contains a parameter, place the query indicator at the end of the last header (VOLTage:PROTection? MAX).

Message Unit Separator

When two or more message units are combined into a compound message, separate the units with a semicolon (STATus:OPERation?;QUESTionable?).

Root Specifier

When it precedes the first header of a message unit, the colon becomes the root specifier. It tells the command parser that this is the root or the top node of the command tree.

Message Terminator

A terminator informs SCPI that it has reached the end of a message. Three permitted messages terminators are:

- ◆ newline (<NL>), which is ASCII decimal 10 or hex 0A.
- ◆ end or identify (<END>)
- ◆ both of the above (<NL><END>).

In the examples of this guide, there is an assumed message terminator at the end of each message.

NOTE: All RS-232 response data sent by the dc source is terminated by the ASCII character pair <carriage return><newline>. This differs from GPIB response data which is terminated by the single character <newline> with EOI asserted.

SCPI Data Formats

All data programmed to or returned from the dc source is ASCII. The data may be numerical or character string.

Numerical Data Formats

Symbol	Data Form
	<u>Talking Formats</u>
<NR1>	Digits with an implied decimal point assumed at the right of the least-significant digit. Examples: 273
<NR2>	Digits with an explicit decimal point. Example: .0273
<NR3>	Digits with an explicit decimal point and an exponent. Example: 2.73E+2
	<u>Listening Formats</u>
<Nrf>	Extended format that includes <NR1>, <NR2> and <NR3>. Examples: 273 273. 2.73E2
<Nrf+>	Expanded decimal format that includes <Nrf> and MIN MAX . Examples: 273 273. 2.73E2 MAX. MIN and MAX are the minimum and maximum limit values that are implicit in the range specification for the parameter.
<Bool>	Boolean Data. Example: 0 1 or ON OFF

Suffixes and Multipliers

Class	Suffix	Unit	Unit with Multiplier
Current	A	ampere	MA (milliampere)
Amplitude	V	volt	MV (millivolt)
Time	S	second	MS (millisecond)
Common Multipliers			
	1E3	K	kilo
	1E-3	M	milli
	1E-6	U	micro

Response Data Types

Character strings returned by query statements may take either of the the following forms, depending on the length of the returned string:

- <CRD> Character Response Data. Permits the return of character strings.
- <AARD> Arbitrary ASCII Response Data. Permits the return of undelimited 7-bit ASCII. This data type has an implied message terminator.
- <SRD> String Response Data. Returns string parameters enclosed in double quotes.

SCPI Command Completion

SCPI commands sent to the dc source are processed either sequentially or in parallel. Sequential commands finish execution before a subsequent command begins. Parallel commands allow other commands to begin executing while the parallel command is still executing. Commands that affect trigger actions are among the parallel commands.

The `*WAI`, `*OPC`, and `*OPC?` common commands provide different ways of indicating when all transmitted commands, including any parallel ones, have completed their operations. The syntax and parameters for these commands are described in chapter 4. Some practical considerations for using these commands are as follows:

<code>*WAI</code>	This prevents the dc source from processing subsequent commands until all pending operations are completed.
<code>*OPC?</code>	This places a 1 in the Output Queue when all pending operations have completed. Because it requires your program to read the returned value before executing the next program statement, <code>*OPC?</code> can be used to cause the controller to wait for commands to complete before proceeding with its program.
<code>*OPC</code>	This sets the OPC status bit when all pending operations have completed. Since your program can read this status bit on an interrupt basis, <code>*OPC</code> allows subsequent commands to be executed.

NOTE: The trigger subsystem must be in the Idle state in order for the status OPC bit to be true. Therefore, as far as triggers are concerned, OPC is false whenever the trigger subsystem is in the Initiated state.

Using Device Clear

You can send a device clear at any time abort a SCPI command that may be hanging up the GPIB interface. The status registers, the error queue, and all configuration states are left unchanged when a device clear message is received. Device clear performs the following actions:

- ◆ The input and output buffers of the dc source are cleared.
- ◆ The dc source is prepared to accept a new command string.

The following statement shows how to send a device clear over the GPIB interface using *Keysight BASIC*:

```
CLEAR 705      IEEE-488 Device Clear
```

The following statement shows how to send a device clear over the GPIB interface using the GPIB command library for *C* or *QuickBASIC*:

```
IOCLEAR (705)
```

NOTE: For RS-232 operation, sending a Break will perform the same operation as the IEE-488 device clear message.

Programming the DC Source

Introduction

This chapter contains examples on how to program your dc source. Simple examples show you how to program:

- ◆ output functions such as voltage and current
- ◆ internal and external triggers
- ◆ measurement functions
- ◆ the status and protection functions

NOTE: These examples in this chapter show which commands are used to perform a particular function, but do not show the commands being used in any particular programming environment. Refer to Appendix D for some examples of SCPI commands in a specific programming environment.

Programming the Output

Power-on Initialization

When the dc source is first turned on, it wakes up with the output state set OFF. In this state the output voltage is set to 0. The following commands are given implicitly at power-on:

```
*RST
*CLS
STATus:PRESet
*SRE 0
*ESE 0
```

*RST is a convenient way to program all parameters to a known state. Refer to the *RST command in chapter 4 to see how each programmable parameter is set by *RST. Refer to the *PSC command in chapter 4 for more information on the power-on initialization of the *ESE and the *SRE registers.

Enabling the Output

To enable the output, use the command:

```
OUTPut ON
```

3 - Programming the DC Source

Output Voltage

The output voltage is controlled with the VOLTage command. For example, to set the output voltage to 25 volts, use:

```
VOLTage 25
```

The dc source can be programmed to turn off its output if the output voltage exceeds a preset peak voltage limit. This protection feature is implemented with the VOLTage:PROTection command as explained in chapter 4.

Maximum Voltage

The maximum rms output voltage that can be programmed can be queried with:

```
VOLTage? MAX
```

Output Current

All models have a programmable current function. The command to program the current is:

```
CURRent <n>
```

where <n> is the current limit in amperes.

If the load attempts to draw more current than the programmed limit, the output voltage is reduced to keep the current within the limit.

Maximum Current

The maximum output current that can be programmed can be queried with:

```
CURRent? MAX
```

Overcurrent Protection

The dc source can also be programmed to turn off its output if the current limit is reached. As explained in chapter 4, this protection feature is implemented the following command:

```
CURRent:PROTection:STAtE ON | OFF
```

NOTE:	Use OUTP:PROT:DEL to prevent momentary current limit conditions caused by programmed output changes from tripping the overcurrent protection.
--------------	---

Triggering Output Changes

The dc source has two independent trigger systems. One is used for generating output changes, and the other is used for triggering measurements. This section describes the output trigger system. The measurement trigger system is described under "Triggering Measurements".

SCPI Triggering Nomenclature

In SCPI terms, trigger systems are called sequences. When more than one trigger system exists, they are differentiated by naming them SEQUENCE1 and SEQUENCE2. SEQUENCE1 is the transient trigger system and SEQUENCE2 is the measurement trigger system. The dc source uses aliases with more descriptive names for these sequences. These aliases can be used instead of the sequence forms.

Sequence Form	Alias
SEQUENCE1	TRANSient
SEQUENCE2	ACQUIRE

Output Trigger System Model

Figure 3-1 is a model of the output trigger system. The rectangular boxes represent states. The arrows show the transition to occur.

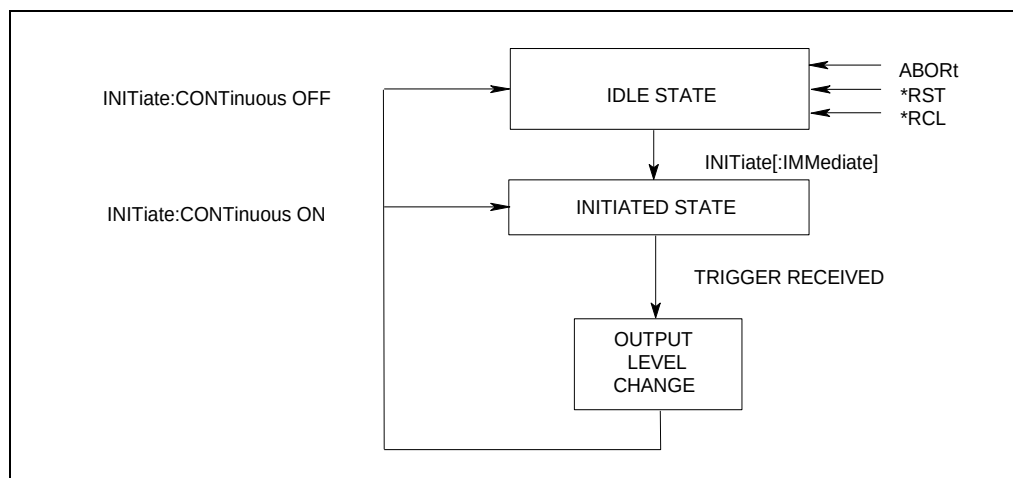


Figure 3-1. Model of Output Triggers

Setting the Voltage or Current Trigger Levels

To program output trigger levels, you must first specify a voltage or current trigger level that the output will go to once a trigger signal is received. Use the following commands to set the output trigger level:

```
VOLTage:TRIGgered <n>    or
CURRent:TRIGgered <n>
```

NOTE: Until they are programmed, uninitialized trigger levels will assume their corresponding immediate levels. For example, if a dc source is powered up and VOLTage:LEVel is programmed to 6, then VOLTage:LEVel:TRIGger will also be 6 until you program it to another value. Once you program VOLTage:LEVel:TRIGger to a value, it will remain at that value regardless of how you subsequently reprogram VOLTage:LEVel.

Initiating the Output Trigger System

When the dc source is turned on, the trigger subsystem is in the idle state. In this state, the trigger subsystem ignores all triggers. Sending the following commands at any time returns the trigger system to the Idle state:

```
ABORt
*RST
*RCL
```

The INITiate commands move the trigger system from the Idle state to the Initiated state. This enables the dc source to receive triggers. To initiate for a single triggered action, use:

```
INITiate:SEQuence1 or
INITiate:NAME TRANsient
```

After a trigger is received and the action completes, the trigger system will return to the Idle state. Thus it will be necessary to initiate the system each time a triggered action is desired.

To keep a trigger system initiated for multiple actions without having to send an initiate command for each trigger, use:

```
INITiate:CONTInuous:SEQuence1 ON      or
INITiate:CONTInuous:NAME TRANsient, ON
```

Generating Triggers

You can only program output triggers over the GPIB bus. Since BUS is the only trigger source for output triggers, the following command is provided for completeness only:

```
TRIGger:SOURce BUS
```

After you have specified the appropriate trigger source, you can generate triggers as follows:

Single Triggers Send one of the following commands over the GPIB:

```
TRIGger:IMMediate
*TRG
a group execute trigger
```

Continuous Triggers Send the following command over the GPIB:

```
INITiate:CONTInuous:SEQuence1 ON
```

When the trigger system enters the Output Change state upon receipt of a trigger (see figure 3-1), the triggered functions are set to their programmed trigger levels. When the triggered actions are completed, the trigger system returns to the Idle state.

Making Measurements

The dc source has the ability to make several types of voltage or current measurements. The measurement capabilities of the Keysight 66312A and Keysight 66332A models are particularly useful for loads that draw current in pulses.

NOTE: You cannot measure output voltage and current simultaneously.

All measurements are performed by digitizing the instantaneous output voltage or current for a defined number of samples and sample interval, storing the results in a buffer, and then calculating the measured result. Many parameters of the measurement are programmable. These include the number of samples, the time interval between samples, the bandwidth, and the method of triggering. Note that there is a tradeoff between these parameters and the speed, accuracy, and stability of the measurement in the presence of noise.

There are two ways to make measurements:

- ◆ Use the MEASure commands to immediately start acquiring new voltage or current data, and return measurement calculations from this data as soon as the buffer is full. This is the easiest way to make measurements, since it requires no explicit trigger programming.
- ◆ Use an acquisition trigger to acquire the data. Then use the FETCh commands to return calculations from the data that was retrieved by the acquisition trigger. This method gives you the flexibility to synchronize the data acquisition with a transition in the output voltage or current. FETCh commands do not trigger the acquisition of new measurement data, but they can be used to return many different calculations from the data that was retrieved by the acquisition trigger. Note that if you take a voltage measurement, you can fetch only voltage data.

Making triggered measurements with the acquisition trigger system is discussed under "Triggering Measurements".

NOTE: For each MEASure form of the query, there is a corresponding query that begins with the header FETCh. FETCh queries perform the same calculation as their MEASure counterparts, but do not cause new data to be acquired. Data acquired by an explicit trigger or a previously programmed MEASure command are used.

Voltage and Current Measurements

The SCPI language provides a number of MEASure and FETCh queries which return various measurement parameters of voltage and current waveforms.

DC Measurements

To measure the dc output voltage or current, use:

```
MEASure:VOLTage? or
MEASure:CURREnt?
```

Dc voltage and current is measured by acquiring a number of readings at the selected time interval, applying a Hanning window function to the readings, and averaging the readings. Windowing is a signal conditioning process that reduces the error in dc measurements made in the presence of periodic signals such as line ripple. At power-on and after a *RST command, the following parameters are set:

```
SENSe:SWEep:TINTerval 15.6E-6
SENSe:SWEep:POINTs 2048
```


3 - Programming the DC Source

This results in a data acquisition time of 32 milliseconds. Adding a command processing overhead of about 20 milliseconds results in a total measurement time of about 50 milliseconds per measurement sample.

Ripple rejection is a function of the number of cycles of the ripple frequency contained in the acquisition window. More cycles in the acquisition window results in better ripple rejection. If you increase the time interval for each measurement to 45 microseconds for example, this results in 5.53 cycles in the acquisition window at 60 Hz, for a ripple rejection of about 70 dB.

Note that the speed of the measurement can be increased by reducing the number of sample points. For example, the commands

```
SENSe:SWEep:TINterval 15E-6  
SENSe:SWEep:POINts 1024
```

speeds up the acquisition period to 16 milliseconds; however, the tradeoff is reduced measurement accuracy.

RMS Measurements (Keysight 66312A, 66332A Only)

To read the rms content of a voltage or current waveform, use:

```
MEASure:VOLTage:ACDC? or  
MEASure:CURREnt:ACDC?
```

This returns the total rms measurement, including the dc portion.

Making rms measurements on ac waveforms for which a non-integral number of cycles of data has been acquired may result in measurement errors due to the last partial cycle of acquired data. The instrument reduces this error by using a Hanning window function when making the measurement.

Minimum and Maximum Measurements (Keysight 66312A, 66332A Only)

To measure the maximum or minimum voltage or current of a pulse or ac waveform, use:

```
MEASure:VOLTage:MAXimum?  
MEASure:VOLTage:MINimum?  
MEASure:CURREnt:MAXimum?  
MEASure:CURREnt:MINimum?
```

Current Ranges

The dc source has two current measurement ranges. The command that controls the ranges is:

```
SENSe:CURREnt:RANGe MIN | MAX
```

When the range is set to MIN, the maximum current that can be measured is 20 milliamperes.

Returning Measurement Data From the Data Buffer (Keysight 66312A, 66332A Only)

The MEASure and FETCh queries can also return all data values of the instantaneous voltage or current buffer. The commands are:

```
MEASure:ARRay:CURREnt?  
MEASure:ARRay:VOLTage?
```

Internally Triggered Measurements

You can use the data acquisition trigger system to synchronize the timing of the voltage and current data acquisition with a BUS or internal trigger source. Then use the FETCH commands to return different calculations from the data acquired by the measurement trigger.

SCPI Triggering Nomenclature

As previously explained under "Triggering Output Changes", the dc source uses the following sequence name and alias for the measurement trigger system. This alias can be used instead of the sequence form.

Sequence Form	Alias
SEQUence2	ACQuire

Measurement Trigger System Model

Figure 3-2 is a model of the measurement trigger system. The rectangular boxes represent states. The arrows show the transitions between states. These are labeled with the input or event that causes the transition to occur.

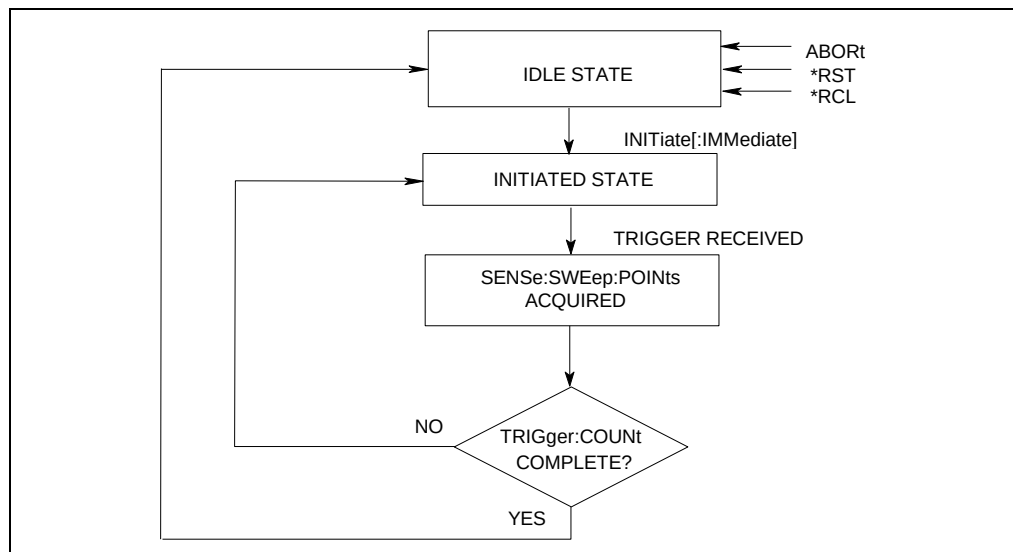


Figure 3-2. Model of Measurement Triggers

Initiating the Measurement Trigger System (Keysight 66312A, 66332A Only)

When the dc source is turned on, the trigger system is in the idle state. In this state, the trigger system ignores all triggers. Sending the following commands at any time returns the trigger system to the Idle state:

```

ABORt
*RST
*RCL

```

The INITiate commands move the trigger system from the Idle state to the Initiated state. This enables the dc source to receive triggers. To initiate for a measurement trigger, use:

3 - Programming the DC Source

```
INITiate:SEquence2 or  
INITiate:NAME ACquire
```

After a trigger is received and the data acquisition completes, the trigger system will return to the Idle state (unless multiple measurements are desired). Thus it will be necessary to initiate the system each time a triggered acquisition is desired.

NOTE: You cannot initiate measurement triggers continuously. Otherwise, the measurement data in the data buffer would continuously be overwritten by each triggered measurement.

Selecting the Measurement Trigger Source (Keysight 66312A, 66332A Only)

The trigger system is waiting for a trigger signal in the Initiated state. Before you generate a trigger, you must select a trigger source. The following measurement trigger sources can be selected:

BUS - selects GPIB bus triggers.
INTernal - selects the dc source's output as the measurement trigger.

To select GPIB bus triggers (group execute trigger, device trigger, or *TRG command), use:

```
TRIGger:SEquence2:SOURce BUS or  
TRIGger:ACquire:SOURce BUS
```

To select internal triggers (measurements triggered off the output signal) use:

```
TRIGger:SEquence2:SOURce INTernal or  
TRIGger:ACquire:SOURce INTernal
```

Generating Measurement Triggers (Keysight 66312A, 66332A Only)

There is only one measurement converter in the dc source. Before you generate a measurement trigger, you must specify a measurement acquisition of either voltage or current. To specify a measurement acquisition use:

```
SENSe:FUNCTion "CURRent" or  
SENSe:FUNCTion "VOLTage"
```

Providing that you have specified the appropriate trigger source and a measurement acquisition, you can generate triggers as follows:

GPIB Triggers	Send one of the following commands over the GPIB: TRIGger:IMMediate (not affected by the trigger source setting) *TRG a group execute trigger
Internal Triggers	To trigger off of the output signal, you must specify the output level that generates the trigger, the rising or falling edge of the slope, and a hysteresis to qualify trigger conditions. This is illustrated in figure 3-3.

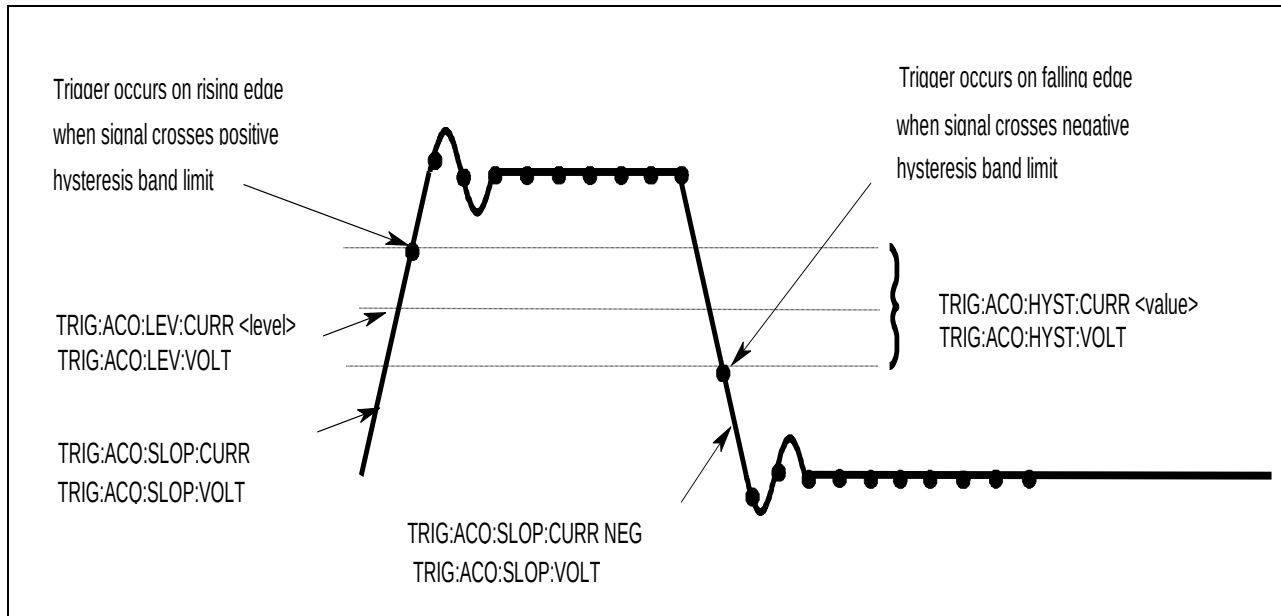


Figure 3-3. Trigger Commands Used to Measure Output Pulses

To specify the output level that will generate triggers for both positive- and negative-going signals use:

```
TRIGger:SEquence2:LEVel:CURRent <value> or
TRIGger:ACQuire:LEVel:CURRent <value>
```

To specify the slope on which triggering occurs use the following commands. You can specify a POSitive, a NEGative, or EITHER type of slope.

```
TRIGger:SEquence2:SLOPe:CURRent <slope> or
TRIGger:ACQuire:SLOPe:CURRent <slope>
```

To specify a hysteresis band to qualify the positive- or negative-going signal use:

```
TRIGger:SEquence2:HYSTeresis:CURRent <value> or
TRIGger:ACQuire:HYSTeresis:CURRent <value>
```

NOTE: When using internal triggers, do not INITiate the measurement until after you have specified the slope, level, and hysteresis.

When the acquisition finishes, any of the FETCH queries can be used to return the results. Once the measurement trigger is initiated, if a FETCH query is sent before the data acquisition is triggered or before it is finished, the response data will be delayed until the trigger occurs and the acquisition completes. This may tie up the controller if the trigger condition does not occur immediately.

One way to wait for results without tying up the controller is to use the SCPI command completion commands. For example, you can send the *OPC command after INITialize, then occasionally poll the OPC status bit in the standard event status register for status completion while doing other tasks. You can also set up an SRQ condition on the OPC status bit going true, and do other tasks until an SRQ interrupt occurs.

Measuring Output Pulses (Keysight 66312A, 66332A Only)

Current Detector

Check that the current detector is set to ACDC when measuring current pulses or other waveforms with a frequency content greater than a few kilohertz.

```
SENSe:CURRent:DETEct ACDC
```

Only select DC as the measurement detector if you are making only DC current measurements and you require a measurement offset better than 2mA on the High current measurement range. Note that this selection gives inaccurate results on current waveforms that have ac content.

```
SENSe:CURRent:DETEct DC
```

Pulse Measurement Queries

The dc so
in Figure :

as shown

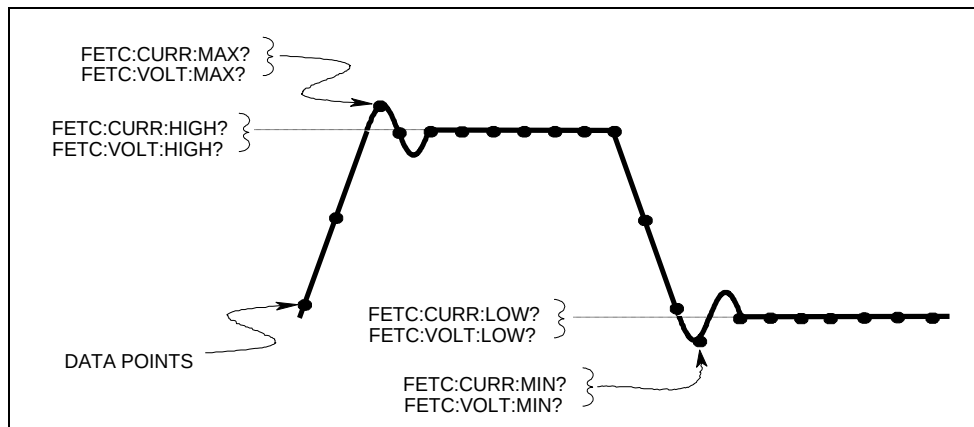


Figure 3-4. Measurement Commands Used to Return Pulse Data

To return the maximum or minimum value of a pulse waveform use the following commands. Note that the data points of the measurement sample may not coincide with the actual maximum or minimum point on the waveform.

```
FETCh:VOLTage:MAXimum? or
FETCh:VOLTage:MINimum?
FETCh:CURRent:MAXimum? or
FETCh:CURRent:MINimum?
```

The average value of the high level or low level of a pulse can also be measured. To return the average value of the high level, use:

```
FETCh:CURRent:HIGH? or
FETCh:VOLTage:HIGH?
```

To return the average value of the low level, use:

```
FETCh:CURRent:LOW? or
FETCh:VOLTage:LOW?
```

Controlling Measurement Samples

Varying

You can vary the number of samples.

Open

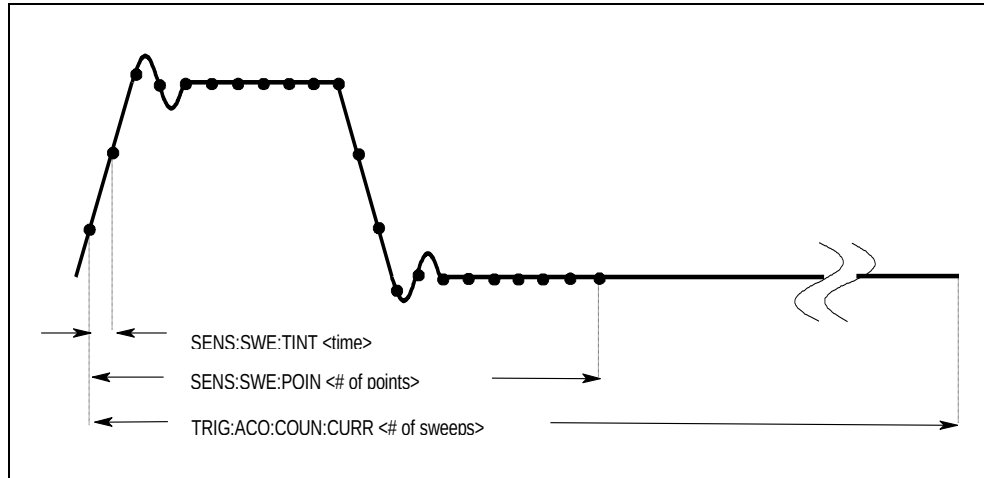


Figure 3-5. Sense Commands Used to Vary the Sampling Rate

At *RST, the output voltage or current sampling rate is 15.6 microseconds. This means that it takes about 32 milliseconds to fill up 2048 data points in the data buffer. You can vary this data sampling rate with:

```
SENSe:SWEep:TINteRval <sample_period>
SENSe:SWEep:POINts <points>
```

For example, to set the time interval to 46.8 microseconds per sample with 1500 samples, use
 SENSe:SWEep:TINteRval 46.8E-6;POINts 1500.

Note that the time interval values are rounded to the nearest 15.6 microsecond increment.

Multiple Measurements (Keysight 66312A, 66332A Only)

The instrument also has the ability to set up several acquisition triggers in succession and average the results from each acquisition in the returned measurement. To set up the trigger system for a number of sequential acquisitions use:

```
TRIGger:ACQuire:COUNt:CURRent <number> or
TRIGger:ACQuire:COUNt:VOLtagE <number>
```

With this setup, the instrument performs each acquisition sequentially, storing the digitized readings in the internal measurement buffer. It is only necessary to initialize the measurement once at the start; after each completed acquisition the instrument will wait for the next valid trigger condition to start another. The results returned by MEASure or FETCh will be the average of the total data acquired.

NOTE: The total number of data points cannot exceed 4096. This means that the product of the trigger count multiplied by the sweep points cannot exceed 4096; otherwise an error will occur.

Pre-event and Post-event Triggering (Keysight 66312A, 66332A Only)

When
voltage
buffer

ut
ition

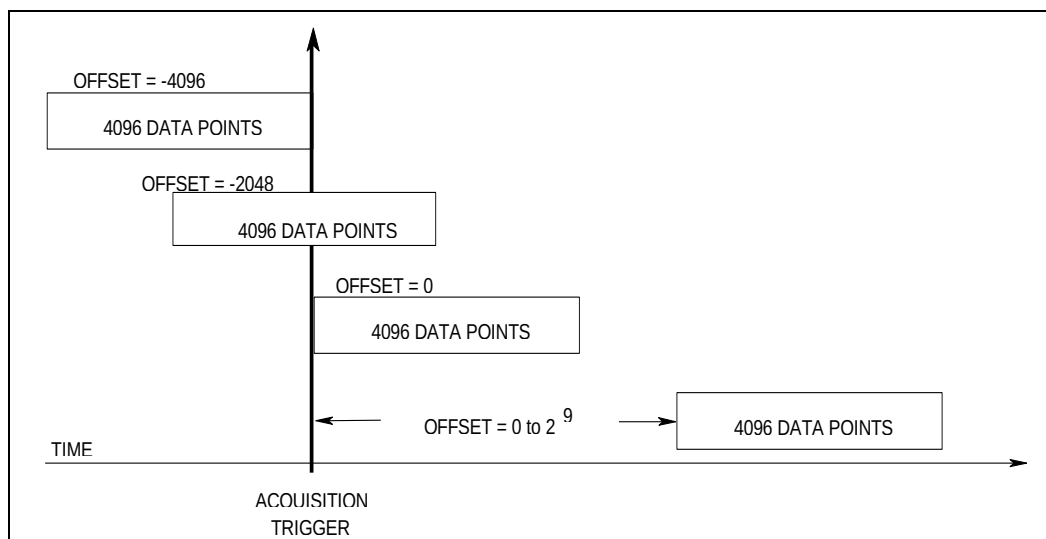


Figure 3-6. Pre-event and Post-event Triggering

To offset the beginning of the acquisition buffer relative to the acquisition trigger, use:

```
SENSe:SWEEp:OFFSet:POINts <offset>
```

The range for the offset is -4096 to 2,000,000,000 points. As shown in the figure, when the offset is negative, the values at the beginning of the data record represent samples taken prior to the trigger. When the value is 0, all of the values are taken after the trigger. Values greater than zero can be used to program a delay time from the receipt of the trigger until the data points that are entered into the buffer are valid. (Delay time = Offset X Sample period)

Pulse Measurement Example (Keysight 66312A, 66332A only)

The following program illustrates how to make a pulse measurement over the GPIB. The measurement function is set to ACDC, which gives the best results for current waveforms that have ac content. The measurement incorporates 100 readings taken at time intervals of 31.2 microseconds, for a total measurement time of 3.12 milliseconds. The trigger point for the pulse measurement occurs at 0.1 amperes on the positive slope of the current pulse. The measurement offset is programmed so that 20 measurement points prior to the trigger are also returned as part of the measurement sample.

Because measurement triggers are initiated by the current pulse, a FETCh command is used to return the measurement data. FETCh commands are also used to return the MAXimum, MINimum, HIGH, and LOW values of the measurement.

NOTE: MEASure commands cannot be used to return data in this example because they always acquire NEW measurement data each time they are used.

The program can be run on any controller operating under Keysight BASIC. To generate output pulses, an electronic load is programmed to generate 3-ampere pulses with a duty cycle of 100 microseconds at 1000 Hz. The power supply address is 705, and the load address is 706. If required, change these parameters in the appropriate statements.

```

10 !Rev A.00.00
20 OPTION BASE 1
30 DIM Curr_array(100)
40 !
50 ASSIGN @Ps TO 705
60 ASSIGN @Ld TO 706
80 OUTPUT @Ps;"*RST" ! Sets supply to default values
90 OUTPUT @Ps;"OUTP ON" ! Turn on power supply output
100 OUTPUT @Ps;"VOLT 5;CURR 5" ! Program power supply to 5 volts, 5 amps
110 !
120 OUTPUT @Ld;"CURR:LEVEL 0" ! Set up electronic load to produce pulses
130 OUTPUT @Ld;"CURR:TLEVEL 3"
140 !
150 OUTPUT @Ld;"TRAN:FREQ 1000"
160 OUTPUT @Ld;"TRAN:DCYCLE 10"
170 OUTPUT @Ld;"TRAN:MODE CONT"
180 OUTPUT @Ld;"TRAN:STATE ON"
190 !
200 OUTPUT @Ps;"SENS:CURR:DET ACDC" ! Set meter to ACDC
210 OUTPUT @Ps;"SENS:CURR:RANG MAX" ! High Current range
220 OUTPUT @Ps;"TRIG:ACQ:SOUR INT" ! Set to trigger on pulse
230 OUTPUT @Ps;"SENS:FUNC ""CURR"" ! Acquire current reading
240 OUTPUT @Ps;"TRIG:ACQ:LEV:CURR .1" ! Trigger at 0.1 amps
250 OUTPUT @Ps;"TRIG:ACQ:SLOPE:CURR POS" ! Trigger on positive slope
260 OUTPUT @Ps;"TRIG:ACQ:HYST:CURR .05" ! Set hysteresis of trigger
270 OUTPUT @Ps;"SENS:SWE:TINT 30E-6" ! Time interval of 30us will be rounded up to 31.2us
280 OUTPUT @Ps;"SENS:SWE:POIN 100" ! Set number of measurement samples in sweep
290 OUTPUT 705;"SENS:SWE:OFFS:POIN -20" ! Number of sample points before trigger
300 OUTPUT @Ps;"INIT:NAME ACQ" ! Initiate the trigger system.
310 ! Controller now waits for trigger to occur.
320 OUTPUT @Ps;"FETCH:ARRAY:CURR?" ! Get the data after measurement completes.
330 !
340 ENTER @Ps;Curr_array(*) ! Enters all 100 data points
350 PRINT Curr_array(*) ! Print all data points
360 !
370 OUTPUT @Ps;"FETCH:CURR:MAX?" ! Get more data from previous measurement.
380 ENTER @Ps;Curr_max
390 PRINT "MAX CURRENT",Curr_max
400 !
410 OUTPUT @Ps;"FETCH:CURR:MIN?"
420 ENTER @Ps;Curr_min
430 PRINT "MIN CURRENT",Curr_min
440 !
450 OUTPUT @Ps;"FETCH:CURR:HIG?"
460 ENTER @Ps;Curr_hi
470 PRINT "HIGH CURRENT",Curr_hi
480 !
490 OUTPUT @Ps;"FETCH:CURR:LOW?"
500 ENTER @Ps;Curr_low
510 PRINT "LOW CURRENT",Curr_low
520 !
530 END

```

When this program runs, it returns 100 measurement data points as well as the MIN, MAX, HIGH, and LOW data in the following format:

.030585	.031869	.0344369	.031655	.0320829	.0325109	.0333669	.0340089
.0320825	.031449	.031227	.031441	.0337949	.0327249	.031869	.031655
.0327249	.031013	.0325109	.0333669	3.09751	3.1814	3.14266	3.13667
3.13817	3.13624	.977283	.0667496	.0245932	.0280171	.031013	.031655
.0331529	.0350788	.0348648	.0327249	.031227	.0327249	.031227	.030799
.031869	.0329389	.030371	.031655	.031869	.0329389	.031869	.0322869
.0320829	.0325109	.0333669	.0340089	.0348648	.0327249	.031227	.0327249
.0320829	.030371	.031449	.031227	.031441	.0337949	.031449	.0333669
.031441	.0337949	.030371	.031655	.031869	.0329389	.031869	.0293011
.031441	.0337949	.0327249	.031869	.031655	.031655	.0320829	.031227
.0322969	.031655	.0327249	.0340089	2.97661	3.18632	3.14523	3.13496
3.13453	3.13731	1.32438	.0836549	.0258772	.0284451	.0275891	.0329389
.0329389	.0333669	.0322969	.0333669				
MAX CURRENT		3.18632					
MIN CURRENT		.0245932					
HIGH CURRENT		3.1371					
LOW CURRENT		.0314077					

Programming the Status Registers

You can use status register programming to determine the operating condition of the dc source at any time. For example, you may program the dc source to generate an interrupt (assert SRQ) when an event such as a current limit occurs. When the interrupt occurs, your program can then act on the event in the appropriate fashion.

Figure 3-7 shows the status register structure of the dc source. Table 3-1 defines the status bits. The Standard Event, Status Byte, and Service Request Enable registers and the Output Queue perform standard GPIB functions as defined in the *IEEE 488.2 Standard Digital Interface for Programmable Instrumentation*. The Operation Status and Questionable Status registers implement functions that are specific to the dc source.

Power-On Conditions

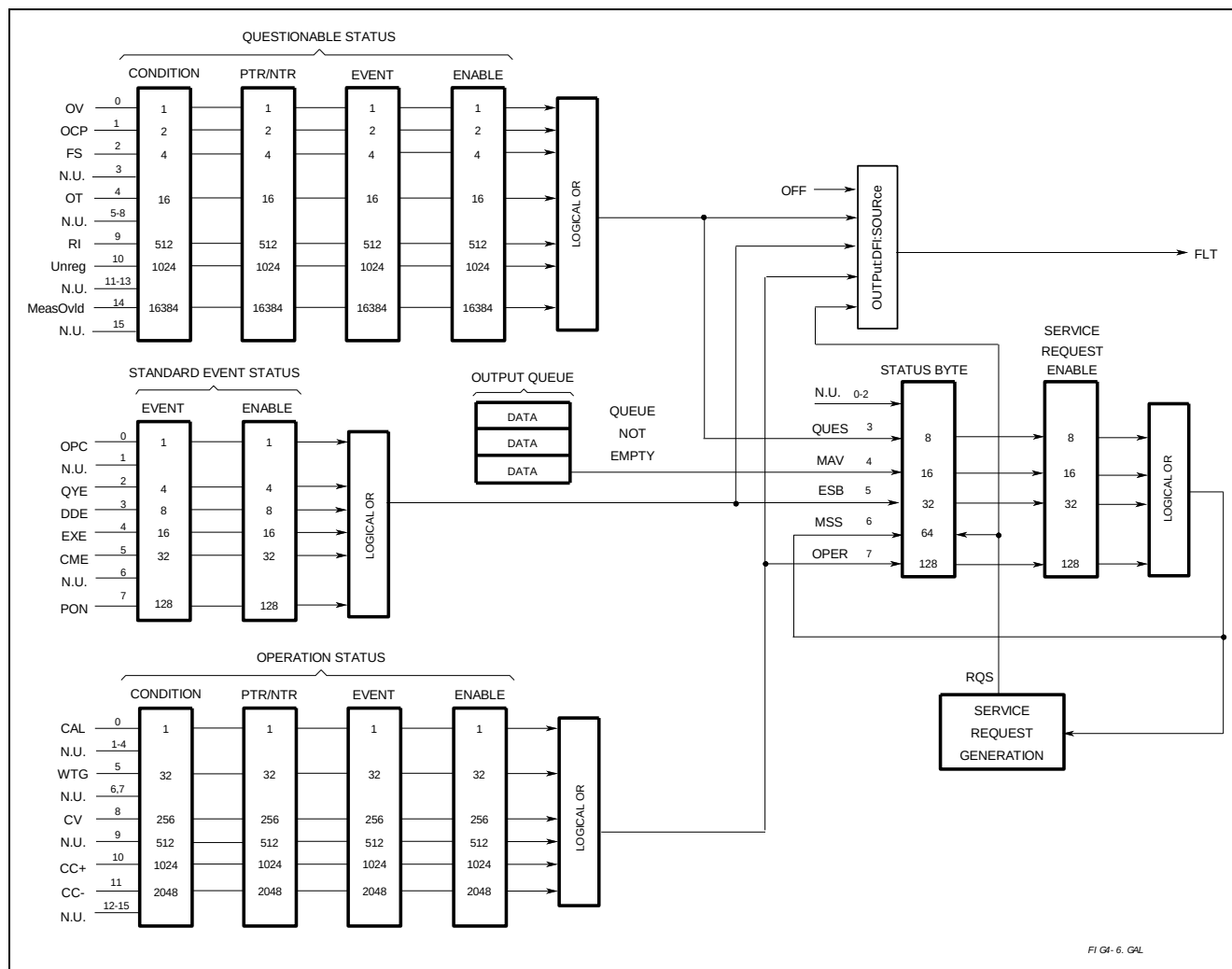


Table 3-1. Bit Configurations of Status Registers

Bit	Signal	Meaning
0	CAL	Operation Status Group The dc source is computing new calibration constants
5	WTG	The dc source is waiting for a trigger
8	CV	The dc source is in constant voltage mode
10	CC+	The dc source is in constant current mode
11	CC-	The dc source is in negative constant current mode
0	OV	Questionable Status Group The overvoltage protection has tripped
1	OCP	The overcurrent protection has tripped
2	FS	The fuse is blown
4	OT	The overtemperature protection has tripped
9	RI	The remote inhibit state is active
10	Unreg	The output is unregulated
14	MeasOvld	Current measurement exceeded capability of low range
0	OPC	Standard Event Status Group Operation complete
2	QYE	Query error
3	DDE	Device-dependent error
4	EXE	Execution error
5	CME	Command error
7	PON	Power-on
3	QUES	Status Byte and Service Request Enable Registers Questionable status summary bit
4	MAV	Message Available summary bit
5	ESB	Event Status Summary bit
6	MSS	Master Status Summary bit
	RQS	Request Service bit
7	OPER	Operation status summary bit

Operation Status Group

The Operation Status registers record signals that occur during normal operation. As shown below, the group consists of a Condition, PTR/NTR, Event, and Enable register. The outputs of the Operation Status register group are logically-ORed into the OPER(ation) summary bit (7) of the Status Byte register.

Register	Command	Description
Condition	STAT:OPER:COND?	A register that holds real-time status of the circuits being monitored. It is a read-only register.
PTR Filter	STAT:OPER:PTR <n>	A positive transition filter that functions as described under STAT:OPER:NTR PTR commands in chapter 4. It is a read/write register.
NTR Filter	STAT:OPER:NTR <n>	A negative transition filter that functions as described under STAT:OPER:NTR PTR commands in chapter 4. It is a read/write register.
Event	STAT:OPER:EVEN?	A register that latches any condition that is passed through the PTR or NTR filters. It is a read-only register that is cleared when read.
Enable	STAT:OPER:ENAB <n>	A register that functions as a mask for enabling specific bits from the Event register. It is a read/write register.

Questionable Status Group

The Questionable Status registers record signals that indicate abnormal operation of the dc source. As shown in figure 3-7, the group consists of the same type of registers as the Status Operation group. The outputs of the Questionable Status group are logically-ORed into the QUESTionable summary bit (3) of the Status Byte register.

Register	Command	Description
Condition	STAT:QUES:COND?	A register that holds real-time status of the circuits being monitored. It is a read-only register.
PTR Filter	STAT:QUES:PTR <n>	A positive transition filter that functions as described under STAT:QUES:NTR PTR commands in chapter 4. It is a read/write register.
NTR Filter	STAT:QUES:NTR <n>	A negative transition filter that functions as described under STAT:QUES:NTR PTR commands in chapter 4. It is a read/write register.
Event	STAT:QUES:EVEN?	A register that latches any condition that is passed through the PTR or NTR filters. It is a read-only register that is cleared when read.
Enable	STAT:QUES:ENAB <n>	A register that functions as a mask for enabling specific bits from the Event register. It is a read/write register..

Standard Event Status Group

This group consists of an Event register and an Enable register that are programmed by Common commands. The Standard Event event register latches events relating to instrument communication status (see figure 3-7). It is a read-only register that is cleared when read. The Standard Event enable register functions similarly to the enable registers of the Operation and Questionable status groups.

Command	Action
*ESE	programs specific bits in the Standard Event enable register.
*PSC ON	clears the Standard Event enable register at power-on.
*ESR?	reads and clears the Standard Event event register.

The PON (Power On) Bit

The PON bit in the Standard Event event register is set whenever the dc source is turned on. The most common use for PON is to generate an SRQ at power-on following an unexpected loss of power. To do this, bit 7 of the Standard Event enable register must be set so that a power-on event registers in the ESB (Standard Event Summary Bit), bit 5 of the Service Request Enable register must be set to permit an SRQ to be generated, and *PSC OFF must be sent. The commands to accomplish these conditions are:

```
*PSC OFF    *ESE 128    *SRE 32
```

Status Byte Register

This register summarizes the information from all other status groups as defined in the *IEEE 488.2 Standard Digital Interface for Programmable Instrumentation*. The bit configuration is shown in Table 3-1.

Command	Action
*STB?	reads the data in the register but does not clear it (returns MSS in bit 6)
serial poll	clears RQS inside the register and returns it in bit position 6 of the response.

The MSS Bit

This is a real-time (unlatched) summary of all Status Byte register bits that are enabled by the Service Request Enable register. MSS is set whenever the dc source has one or more reasons for requesting service. *STB? reads the MSS in bit position 6 of the response but does not clear any of the bits in the Status Byte register.

The RQS Bit

The RQS bit is a latched version of the MSS bit. Whenever the dc source requests service, it sets the SRQ interrupt line true and latches RQS into bit 6 of the Status Byte register. When the controller does a serial poll, RQS is cleared inside the register and returned in bit position 6 of the response. The remaining bits of the Status Byte register are not disturbed.

The MAV Bit and Output Queue

The Output Queue is a first-in, first-out (FIFO) data register that stores dc source-to-controller messages until the controller reads them. Whenever the queue holds one or more bytes, it sets the MAV bit (4) of the Status Byte register.

Determining the Cause of a Service Interrupt

You can determine the reason for an SRQ by the following actions:

- | | |
|--------|---|
| Step 1 | Determine which summary bits are active. Use:
*STB? or serial poll |
| Step 2 | Read the corresponding Event register for each summary bit to determine which events caused the summary bit to be set. Use:
STATUS:QUESTIONABLE:EVENT?
STATUS:OPERATION:EVENT?
ESR?
When an Event register is read, it is cleared. This also clears the corresponding summary bit. |
| Step 3 | Remove the specific condition that caused the event. If this is not possible, the event may be disabled by programming the corresponding bit of the status group Enable register or NTR PTR filter. A faster way to prevent the interrupt is to disable the service request by programming the appropriate bit of the Service Request Enable register |

Servicing Operation Status and Questionable Status Events

This example assumes you want a service request generated whenever the dc source switches to the CC (constant current) operating mode, or whenever the dc source's overvoltage, overcurrent, or overtemperature circuits have tripped. From figure 3-7, note the required path for a condition at bit 10 (CC) of the Operation Status register to set bit 6 (RQS) of the Status Byte register. Also note the required path for Questionable Status conditions at bits 0, 1, and 4 to generate a service request (RQS) at the Status Byte register. The required register programming is as follows:

- | | |
|--------|---|
| Step 1 | Program the Operation Status PTR register to allow a positive transition at bit 10 to be latched into the Operation Status Event register, and allow the latched event to be summed into the Operation summary bit. Use:
STATUS:OPERATION:PTR 1024;ENABLE 1024 |
| Step 2 | Program the Questionable Status PTR register to allow a positive transition at bits 0, 1, or 4 to be latched into the Questionable Status Event register, and allow the latched event to be summed into the Questionable summary bit. Use: |

3 - Programming the DC Source

```
STATUS:QUESTIONABLE:PTR 19;ENABLE 19 (1 + 2 + 16 = 19)
```

Step 3 Program the Service Request Enable register to allow both the Operation and the Questionable summary bits from the Status Byte register to generate RQS. Use:
*SRE 136 (8 + 128 = 136)

Step 4 When you service the request, read the event registers to determine which Operation Status and Questionable Status Event register bits are set, and clear the registers for the next event. Use:
STATUS:OPERation:EVENT;QUESTIONABLE:EVENT?

Monitoring Both Phases of a Status Transition

You can monitor a status signal for both its positive and negative transitions. For example, to generate RQS when the dc source either enters the CC+ (constant current) condition or leaves that condition, program the Operational Status PTR/NTR filter as follows:

```
STATUS:OPERational:PTR 1024;NTR 1024  
STATUS:OPERational:ENABLE 1024;*SRE 128
```

The PTR filter will cause the OPERational summary bit to set RQS when CC+ occurs. When the controller subsequently reads the event register with STATUS:OPERational:EVENT?, the register is cleared. When CC+ subsequently goes false, the NTR filter causes the OPERational summary bit to again set RQS.

Inhibit/Fault Indicator

The remote inhibit(INH) and discrete fault(FLT) indicators are implemented through the respective **INH** and **FLT** connections on the rear panel. Refer to Table 1-2 for the electrical parameters.

Remote Inhibit (RI)

Remote inhibit is an external, chassis-referenced logic signal routed through the rear panel INH connection, which allows an external device to signal a fault. To select an operating modes for the remote inhibit signal, use:

```
OUTPut:RI:MODE LATCHing | LIVE | OFF
```

Discrete Fault Indicator (DFI)

The discrete fault indicator is an open-collector logic signal connected to the rear panel FLT connection, that can be used to signal external devices when a fault condition is detected. To select the internal fault source that drives this signal, use:

```
OUTPut:DFI:SOURce QUESTIONABLE | OPERATION | ESB | RQS | OFF
```

To enable or disable the DFI output, use:

```
OUTPut:DFI:STATE ON | OFF
```

Using the Inhibit/Fault Port as a Digital I/O

You can configure the inhibit/fault port to provide a digital input/output to be used with custom digital interface circuits or relay circuits. As shipped from the factory, the port is shipped for inhibit/fault operation. You can change the configuration of the port to operate as a general purpose digital input output port with the following command:

```
[SOURce:]DIGital:FUNCTion RIDFi | DIGio
```

The following table shows the bin assignments of the mating plug when used in RI/DFI mode as well as Digital I/O mode. Refer to Table 1-2 for the electrical characteristics of the port.

Pin	FAULT/INHIBIT	DIGITAL I/O	Bit Weight
1	FLT Output	OUT 0	0
2	FLT Output	OUT 1	1
3	INH Input	IN/OUT 2	2
4	INH Common	Common	not programmable

To program the digital I/O port use:

```
[SOURce:]DIGital:DATA <data>
```

where the data is an integer from 0 to 7 that sets pins 1 to 3 according to their binary weight. Refer to the DIGital:DATA command for more information.

DFI Programming Example

The following program illustrates how to program the DFI port so that it goes low when an OCP condition turns off the output of the unit. To clear an overcurrent condition, the cause of the condition must first be removed and then an OUTput:PROTection:CLEar command must be sent. Note that the status event register will not clear the DFI port until the register is read.

```

10  !Rev A.00.00
20  ASSIGN @Ps TO 705
30  OUTPUT @Ps;"*RST"                ! Sets supply to default values
40  OUTPUT @Ps;"OUTP ON"              ! Turn on power supply output
50  OUTPUT @Ps;"VOLT 10;CURR .1"      ! Program power supply voltage and current
60  !
70  OUTPUT @Ld;"CURR:PROT:STAT ON"    ! Turn on overcurrent protection
80  OUTPUT @Ld;"OUTP:DFI:STAT ON"     ! Turn on DFI port
90  OUTPUT @Ld;"OUTP:DFI:SOUR QUES"    ! Select DFI bit from Questionable status register
100 OUTPUT @Ld;"STAT:QUES:ENAB 2;PTR 2" ! Unmask bit 2 (OCP) on positive transition
110 !
120 OUTPUT @Ld;"OUTP:PROT:CLE"        ! Clears the protection circuit
130 OUTPUT @Ld;"STAT:QUES:EVENT?"     ! Clears the Event register and DFI
140 ENTER @Ld;EVENT                  ! Reads the event and clears the buffer
190 !

```


Language Dictionary

Introduction

This section gives the syntax and parameters for all the IEEE 488.2 SCPI commands and the Common commands used by the dc source. It is assumed that you are familiar with the material in "Chapter 2 - Remote Programming". That chapter explains the terms, symbols, and syntactical structures used here and gives an introduction to programming. You should also be familiar with "Chapter 4 - Front Panel Operation" (in the Operating Guide) in order to understand how the dc source functions.

The programming examples are simple applications of SCPI commands. Because the SCPI syntax remains the same for all programming languages, the examples given for each command are generic.

Syntax Forms	Syntax definitions use the long form, but only short form headers (or "keywords") appear in the examples. Use the long form to help make your program self-documenting.
Parameters	Most commands require a parameter and all queries will return a parameter. The range for a parameter may vary according to the model of dc source. When this is the case, refer to the Specifications table in the Operating Guide.
Models	If a command only applies to specific models, those models are listed in the <Model> Only entry. If there is no <Model> Only entry, the command applies to all models.
Related Commands	Where appropriate, related commands or queries are included. These are listed because they are either directly related by function, or because reading about them will clarify or enhance your understanding of the original command or query.
Order of Presentation	The dictionary is organized according to the following functions: calibration, measurement, output, status, system, and trigger. Both the subsystem commands and the common commands that follow are arranged in alphabetical order under each function.

Subsystem Commands

Subsystem commands are specific to functions. They can be a single command or a group of commands. The groups are comprised of commands that extend one or more levels below the root.

The subsystem command groups are grouped according to function: Calibration, Measurement, Output, Status, System, and Trigger. Commands under each function are grouped alphabetically. Commands followed by a question mark (?) take only the query form. When commands take both the command and query form, this is noted in the syntax descriptions. Table 4-1 lists all of the subsystem commands in alphabetical order.

Table 4-1. Subsystem Commands Syntax

ABORt	Resets the trigger system to the Idle state
CALibrate :	
:CURRent	
[:SOURce]	
[:DC] [:POSitive]	Calibrate positive output current and high current measurement range
:NEGative	Calibrate negative output current
:MEASure	
[:DC] :LOWRange	Calibrate low current measurement range
:AC	Calibrate ac current measurement circuits
:DATA <n>	Input a calibration measurement
:LEVel <level>	Advance to next calibration step (P1 P2)
:PASSword <n>	Set calibration password
:SAVE	Save new cal constants in non-volatile memory
:STATE <bool> [, <n>]	Enable or disable calibration mode
:VOLTage	
[:DC]	Calibrate output voltage and voltage readback
:PROTection	Begin voltage protection calibration sequence
DISPlay	
[:WINDow]	
[:STATE] <bool>	Enable/disable front panel display
:MODE <mode>	Set display mode (NORM TEXT)
:TEXT [:DATA] <string>	Sets the text that is displayed
INITiate	
[:IMMediate]	
:SEQuence[<n>]	Initiate a specific numbered sequence (1 2)
:NAME <name>	Initiate a specific named sequence (TRAN ACQ)
CONTInuous	
:SEQuence1, <bool>	Set continuous initialization
:NAME TRANSient, <bool>	Set continuous initialization
MEASure FETCh	
:ARRay	
:CURRent [:DC]?	Returns the digitized instantaneous current
:VOLTage [:DC]?	Returns the digitized instantaneous voltage
[:SCALar]	
:CURRent [:DC]?	Returns dc current
:ACDC?	Returns the total rms current (ac+dc)
:HIGH?	Returns the HIGH level of a current pulse
:LOW?	Returns the LOW level of a current pulse
:MAX?	Returns maximum current
:MIN?	Returns minimum current
:VOLTage [:DC]?	Returns dc voltage
:ACDC?	Returns the total rms voltage (ac+dc)
:HIGH?	Returns the HIGH level of a voltage pulse
:LOW?	Returns the LOW level of a voltage pulse
:MAX?	Returns maximum voltage
:MIN?	Returns minimum voltage

Table 4-1. Subsystem Commands Syntax (continued)

OUTPut	
:STATe <bool> [,NORelay]	Enables/disables the dc source output
:DFI	
[:STATe] <bool>	Enable/disable DFI output
:SOURce <source>	Selects event source (QUES OPER ESB RQS OFF)
:PON	
:STATe <state>	Set power-on state (*RST RCL0)
:PROTection	
:CLEar	Reset latched protection
:DELAy <n>	Delay after programming/before protection
:RELAy	
[:STATe] <bool>	Opens/closes the external relay contacts
:POLarity <polarity>	Sets the external relay polarity (NORM REV)
:RI	
:MODE <mode>	Sets remote inhibit input (LATC LIVE OFF)
SENSe	
:CURRent	
[:DC]	
RANGE [:UPPer] <n>	Selects the high current measurement range
:DETection <detector>	Selects the current measurement detector (ACDC DC)
:FUNction <function>	Configures the measurement sensor ("VOLT" "CURR")
:SWEep	
:OFFSet	
:POINts <n>	Defines the offset in the data sweep
:POINts <n>	Define the number of data points in a sweep
:TINterval <n>	Sets the digitizer sample spacing
:WINDow [:TYPE] <type>	Sets the measurement window function (HANN RECT)
[SOURce:]	
CURRent	
[:LEVel]	
[:IMMediate][:AMPLitude] <n>	Sets the output current level
:TRIGgered [:AMPLitude] <n>	Sets the triggered output current level
:PROTection	
:STATe <bool>	Enable/Disable current limit protection
DIGital	
:DATA [:VALue] <n>	Sets and reads the digital control port
:FUNction <function>	Configures digital control port (RIDF DIG)
VOLTage	
[:LEVel]	
[:IMMediate][:AMPLitude] <n>	Sets the dc voltage level
:TRIGgered [:AMPLitude] <n>	Sets the transient voltage level
:ALC	
:BANDwidth? :BWIDth?	Returns setting of output mode switch
:PROTection [:LEVel] <n>	Sets the overvoltage protection threshold

Table 4-1. Subsystem Commands Syntax (continued)

STATus	
:PRESet	Presets all enable and transition registers to power-on
:OPERation	
[:EVENT]?	Returns the value of the event register
:CONDition?	Returns the value of the condition register
:ENABle <n>	Enables specific bits in the Event register
:NTRansition<n>	Sets the Negative transition filter
:PTRansition<n>	Sets the Positive transition filter
:QUEStionable	
[:EVENT]?	Returns the value of the event register
:CONDition?	Returns the value of the condition register
:ENABle <n>	Enables specific bits in the Event register
:NTRansition<n>	Sets the Negative transition filter
:PTRansition<n>	Sets the Positive transition filter
SYSTem	
:ERRor?	Returns the error number and error string
:LANGuage <language>	Sets the programming language (SCPI COMP)
:VERSion?	Returns the SCPI version number
:LOCAL	Go to local mode (for RS-232 operation)
:REMOte	Go to remote mode (for RS-232 operation)
:RWLock	Go to remote with local lockout (for RS-232 operation)
TRIGger	
:SEQuence2 :ACQuire	
[:IMMediate]	Triggers the measurement immediately
:COUNT	
:CURRent <n>	Sets the number of sweeps per current measurement
:VOLTage <n>	Sets the number of sweeps per voltage measurement
:HYSTeresis	
:CURRent <n>	Qualifies the trigger when measuring current
:VOLTage <n>	Qualifies the trigger when measuring voltage
:LEVel	
:CURRent <n>	Sets the trigger level for measuring current
:VOLTage <n>	Sets the trigger level for measuring voltage
:SLOPe	
:CURRent <slope>	Sets the triggered current slope (POS NEG EITH)
:VOLTage <slope>	Sets the triggered voltage slope (POS NEG EITH)
:SOURce <source>	Sets the trigger source (BUS INT)
[:SEQuence1 :TRANsient]	
[:IMMediate]	Triggers the output immediately
:SOURce <source>	Sets the trigger source (BUS)
:SEQuence1	
:DEFine TRANsient	Sets or queries the SEQ1 name
:SEQuence2	
:DEFine ACQuire	Sets or queries the SEQ2 name

Common Commands

Common commands begin with an * and consist of three letters (command) or three letters and a ? (query). They are defined by the IEEE 488.2 standard to perform common interface functions. Common commands and queries are categorized under System, Status, or Trigger functions and are listed at the end of each group. The dc source responds to the following commands:

Table 4-2. Common Commands Syntax

*CLS	Clear status
*ESE <n>	Standard event status enable
*ESE?	Return standard event status enable
*ESR?	Return event status register
*IDN?	Return instrument identification
*OPC	Enable "operation complete" bit in ESR
*OPC?	Return a "1" when operation complete
*OPT?	Return option number
*PSC <bool>	Power-on status clear state set/reset
*PSC?	Return power-on status clear state
*RCL <n>	Recall instrument state
*RST	Reset
*SAV <n>	Save instrument state
*SRE <n>	Set service request enable register
*SRE?	Return service request enable register
*STB?	Return status byte
*TRG	Trigger
*TST?	Perform selftest, then return result
*WAI	Hold off bus until all device commands done

Programming Parameters

The following table lists the output programming parameters for each model.

Table 4-3. Output Programming Parameters

Parameter	Value					
	66312A	66332A	<u>6631B</u> 6611C	<u>6632B</u> 6612C	<u>6633B</u> 6613C	<u>6634B</u> 6614C
[SOUR:]CURR[:LEV][:IMM] MAX and [SOUR:]CURR[:LEV]:TRIG MAX *RST Current Value	2.0475	5.1188	<u>10.237</u> 5.1188	<u>5.1188</u> 2.0475	<u>2.0475</u> 1.0238	<u>1.0238</u> 0.5118
	10% of MAX value for all models					
[SOUR:]VOLT[:LEV][:IMM]MAX and [SOUR:]VOLT[:LEV]:TRIG MAX *RST Voltage Value	20.475	20.475	8.190	20.475	51.188	102.38
	0 V for all models					
[SOUR:]VOLT:PROT[:LEV] MAX *RST OVP Value	22	22	12	22	55	110
	MAX for all models					
OUTP:PROT:DEL MAX *RST Protection Delay Value	2,147,483.647 seconds for all models 0.08 seconds					
SENS:CURR:RANG *RST Current Range	Low range = 0 – 20 mA for all models High Range = 20 mA – MAX for all models Value MAX for all models					

Calibration Commands

Calibration commands let you:

- ◆ Enable and disable the calibration mode
- ◆ Change the calibration password
- ◆ Calibrate the current and voltage programming and measurement, and store new calibration constants in nonvolatile memory.

NOTE: If calibration mode has not been enabled with CALibrate:STATE, programming the calibration commands will generate an error.

CALibrate:CURRent

This command initiates the calibration of the positive dc output current as well as the high-range current measurement circuit.

Command Syntax	CALibrate:CURRent[:SOURce][:DC][:POSitive]
Parameters	None
Examples	CAL:CURRCAL:CURR:SOUR:DC:POS
Related Commands	CAL:CURR:NEG

CALibrate:CURRent:NEGative

This command initiates the calibration of the negative dc output current.

Command Syntax	CALibrate:CURRent[:SOURce][:DC]:NEGative
Parameters	None
Examples	CAL:CURR:NEGCAL:CURR:SOUR:DC:NEG
Related Commands	CAL:CURR

CALibrate:CURRent:MEASure:LOWRange

This command initiates the calibration of the low-range current measurement circuit.

Command Syntax	CALibrate:CURRent:MEASure[:DC]:LOWRange
Parameters	None
Examples	CAL:CURR:MEAS
Related Commands	CAL:CURR

CALibrate:CURRent:MEASure:AC

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This command initiates the calibration of the high bandwidth (ac) measurement circuit.

Command Syntax	CALibrate:CURRent:MEASure:AC
Parameters	None
Examples	CAL:CURR:MEAS:AC

CALibrate:DATA

This command enters a calibration value that you obtain by reading an external meter. You must first select a calibration level (with CALibrate:LEVel) for the value being entered.

Command Syntax	CALibrate:DATA<NRf>	
Parameters	<external reading>	
Unit	A (amperes)	
Examples	CAL:DATA 3222.3 MA	CAL:DATA 5.000
Related Commands	CAL:STAT CAL:LEV	

CALibrate:LEVel

This command selects the next point in the calibration sequence.

P1: the first calibration point

P2: the second calibration point

Command Syntax	CALibrate:LEVel <point>
Parameters	P1 P2
Examples	CAL:LEV P2

CALibrate:PASSword

This command lets you change the calibration password. A new password is automatically stored in nonvolatile memory and does not have to be stored with CALibrate:SAVE.

If the password is set to 0, password protection is removed and the ability to enter the calibration mode is unrestricted.

Command Syntax	CALibrate:PASScode<NRf>	
Parameters	<model number> (default)	
Examples	CAL:PASS 6812	CAL:PASS 6.1994
Related Commands	CAL:SAV	

CALibrate:SAVE

This command saves any new calibration constants after a calibration procedure has been completed in nonvolatile memory. If CALibrate:STATe OFF is programmed without a CALibrate:SAVE, the previous calibration constants are restored..

Command Syntax	CALibrate:SAVE	
Parameters	None	
Examples	CAL:SAVE	
Related Commands	CAL:PASS	CAL:STAT

CALibrate:STATE

This command enables and disables calibration mode. The calibration mode must be enabled before the will accept any other calibration commands.

The first parameter specifies the enabled or disabled state. The second parameter is the password. It is required if the calibration mode is being enabled and the existing password is not 0. If the password is not entered or is incorrect, an error is generated and the calibration mode remains disabled. The query statement returns only the state, not the password.

NOTE: Whenever the calibration state is changed from enabled to disabled, any new calibration constants are lost unless they have been stored with CALibrate:SAVE.

Command Syntax	CALibrate:STATE<bool>[,<NRf>]
Parameters	0 1 OFF ON [,<password>]
*RST Value	OFF
Examples	CAL:STAT 1,6812 CAL:STAT OFF
Query Syntax	CALibrate:STATE?
Returned Parameters	<NR1>
Related Commands	CAL:PASS CAL:SAVE *RST

CALibrate:VOLTage

This command initiates the calibration of the output voltage and the voltage readback circuit.

Command Syntax	CALibrate:VOLTage[:DC]
Parameters	None
Examples	CAL:VOLT CAL:VOLT:DC

CALibrate:VOLTage:PROTection

This command can calibrates the overvoltage protection (OV) circuit. The dc source automatically performs the calibration. CALibrate:VOLTage:PROTection is a sequential command that takes several seconds to complete.

Command Syntax	CALibrate:VOLTage:PROTection
Parameters	None
Examples	CAL:VOLT:PROT

Measurement Commands

Measurement commands consist of measure and sense commands.

Measure commands measure the output voltage or current. Measurements are performed by digitizing the instantaneous output voltage or current for a defined number of samples and sample interval, storing the results in a buffer, and calculating the measured result. Two types of measurement commands are available: MEASure and FETCh. MEASure triggers the acquisition of new data before returning the reading; FETCh returns a reading computed from previously acquired data. If you take a voltage measurement, you can fetch only voltage data.

- ◆ Use MEASure when the measurement does not need to be synchronized with any other event.
- ◆ Use FETCh when it is important that the measurement be synchronized with either a trigger or with a particular part of the output waveform.

Sense commands control the current measurement range, the bandwidth detector of the , and the data acquisition sequence.

MEASure:ARRay:CURRent?

FETCh:ARRay:CURRent?

Keysight 66312A, 66332A Only

These queries return an array containing the instantaneous output current in amperes. The output voltage or output current are digitized whenever a measure command is given or whenever an acquire trigger occurs. The time interval is set by SENSE:SWEep:TINTerval. The position of the trigger relative to the beginning of the data buffer is determined by SENSE:SWEep:OFFSet. The number of points returned is set by SENSE:SWEep:POINts.

Query Syntax	MEASure:ARRay:CURRent[:DC]?
	FETCh:ARRay:CURRent[:DC]?
Parameters	None
Examples	MEAS:ARR:CURR? FETC:ARR:CURR?
Returned Parameters	<NR3>
Related Commands	SENS:SWE:TINT SENS:SWE:OFFS SENS:SWE:POIN

MEASure:ARRay:VOLTagE?

FETCh:ARRay:VOLTagE?

Keysight 66312A, 66332A Only

These queries return an array containing the instantaneous output voltage in volts. The output voltage or output current are digitized whenever a measure command is given or whenever an acquire trigger occurs. The time interval is set by SENSE:SWEep:TINTerval. The position of the trigger relative to the beginning of the data buffer is determined by SENSE:SWEep:OFFSet. The number of points returned is set by SENSE:SWEep:POINts.

Query Syntax	MEASure:ARRay:VOLTagE[:DC]?
	FETCh:ARRay:VOLTagE[:DC]?
Parameters	None
Examples	MEAS:ARR:VOLT? FETC:ARR:VOLT?
Returned Parameters	<NR3>
Related Commands	SENS:SWE:TINT SENS:SWE:OFFS SENS:SWE:POIN

MEASure:CURRENT? FETCh:CURRENT?

FETCh:CURRENT? applies to Keysight 66312A, 66332A Only

These queries return the dc output current.

Query Syntax	MEASure:[SCALar]:CURRENT[:DC]? FETCh:[SCALar]:CURRENT[:DC]?
Parameters	None
Examples	MEAS:CURR? MEAS:CURR:DC?
Returned Parameters	<NR3>
Related Commands	MEAS:VOLT?

MEASure:CURRENT:ACDC? FETCh:CURRENT:ACDC?

Keysight 66312A, 66332A Only

These queries return the ac+dc rms output current.

Query Syntax	MEASure:[SCALar]:CURRENT:ACDC? FETCh:[SCALar]:CURRENT:ACDC?
Parameters	None
Examples	MEAS:CURR:ACDC? FETC:CURR:ACDC?
Returned Parameters	<NR3>
Related Commands	MEAS:VOLT:ACDC?

MEASure:CURRENT:HIGH? FETCh:CURRENT:HIGH?

Keysight 66312A, 66332A Only

These queries return the High level current of a current pulse waveform. The instrument first measures the minimum and maximum data points of the pulse waveform. It then generates a histogram of the pulse waveform using 1024 bins between the maximum and minimum data points. The bin containing the most data points above the 50% point is the high bin. The average of all the data points in the high bin is returned as the High level. If no high bin contains more than 1.25% of the total number of acquired points, then the maximum value is returned by these queries.

Query Syntax	MEASure[SCALar]:CURRENT:HIGH? FETCh[:SCALar]:CURRENT:HIGH?
Parameters	None
Examples	MEAS:CURR:HIGH? FETC:CURR:HIGH?
Returned Parameters	<NR3>
Related Commands	MEAS:CURR:LOW? CALC:REF:HIGH

MEASure:CURRENT:LOW? FETCh:CURRENT:LOW?

Keysight 66312A, 66332A Only

These queries return the Low level current of a current pulse waveform. The instrument first measures the minimum and maximum data points of the pulse waveform. It then generates a histogram of the pulse waveform using 1024 bins between the maximum and minimum data points. The bin containing the most data points below the 50% point is the low bin. The average of all the data points in the low bin is returned as the Low level. If no low bin contains more than 1.25% of the total number of acquired points, then the minimum value is returned by these queries.

Query Syntax	MEASure[SCALar]:CURRENT:LOW? FETCh[:SCALar]:CURRENT:LOW?	
Parameters	None	
Examples	MEAS:Curr:LOW?	FETC:Curr:LOW?
Returned Parameters	<NR3>	
Related Commands	MEAS:Curr:HIGH?	CALC:REF:LOW

MEASure:CURRENT:MAXimum? FETCh:CURRENT: MAXimum?

Keysight 66312A, 66332A Only

These queries return the maximum output current reading from the measurement sample.

Query Syntax	MEASure[:SCALar]:CURRENT:MAXimum? FETCh[:SCALar]:CURRENT:MAXimum?	
Parameters	None	
Examples	MEAS:Curr:MAX?	FETC:Curr:MAX?
Returned Parameters	<NR3>	
Related Commands	MEAS:Curr:MIN?	

MEASure:CURRENT:MINimum? FETCh:CURRENT:MINimum?

Keysight 66312A, 66332A Only

These queries return the minimum output current reading from the measurement sample.

Query Syntax	MEASure[:SCALar]:CURRENT:MINimum? FETCh[:SCALar]:CURRENT:MINimum?	
Parameters	None	
Examples	MEAS:Curr:MIN?	FETC:Curr:MIN?
Returned Parameters	<NR3>	
Related Commands	MEAS:Curr:MAX?	

MEASure:VOLTage? FETCh:VOLTage?

FETCh:VOLTage? applies to Keysight 66312A, 66332A Only

These queries return the dc output voltage.

Query Syntax	MEASure[:SCALar]:VOLTage[:DC]? MEASure[:SCALar]:VOLTage[:DC]?
Parameters	None
Examples	MEAS:VOLT? FETC:VOLT:DC?
Returned Parameters	<NR3>
Related Commands	MEAS:CURRE?

MEASure:VOLTage:ACDC? FETCh:VOLTage:ACDC?

Keysight 66312A, 66332A Only

These queries return the ac+dc rms output voltage.

Query Syntax	MEASure[:SCALar]:VOLTage:ACDC? FETCh[:SCALar]:VOLTage:ACDC?
Parameters	None
Examples	MEAS:VOLT:ACDC? FETC:VOLT:ACDC?
Returned Parameters	<NR3>
Related Commands	MEAS:CURRE:ACDC?

MEASure:VOLTage:HIGH? FETCh:VOLTage:HIGH?

Keysight 66312A, 66332A Only

These queries return the High level voltage of a voltage pulse waveform. The instrument first measures the minimum and maximum data points of the pulse waveform. It then generates a histogram of the pulse waveform using 1024 bins between the maximum and minimum data points. The bin containing the most data points above the 50% point is the high bin. The average of all the data points in the high bin is returned as the High level. If no high bin contains more than 1.25% of the total number of acquired points, then the maximum value is returned by these queries.

Query Syntax	MEASure[SCALar]:VOLTage:HIGH? FETCh[:SCALar]:VOLTage:HIGH?
Parameters	None
Examples	MEAS:VOLT:HIGH? FETC:VOLT:HIGH?
Returned Parameters	<NR3>
Related Commands	MEAS:VOLT:LOW? CALC:REF:HIGH

MEASure:VOLTage:LOW? FETCh:VOLTage:LOW?

Keysight 66312A, 66332A Only

These queries return the Low level voltage of a voltage pulse waveform. The instrument first measures the minimum and maximum data points of the pulse waveform. It then generates a histogram of the pulse waveform using 1024 bins between the maximum and minimum data points. The bin containing the most data points below the 50% point is the low bin. The average of all the data points in the low bin is returned as the Low level. If no low bin contains more than 1.25% of the total number of acquired points, then the minimum value is returned by these queries.

Query Syntax	MEASure[SCALar]:VOLTage:LOW? FETCh[:SCALar]:VOLTage:LOW?	
Parameters	None	
Examples	MEAS:VOLT:LOW?	FETC:VOLT:LOW?
Returned Parameters	<NR3>	
Related Commands	MEAS:VOLT:HIGH?	CALC:REF:LOW

MEASure:VOLTage:MAXimum? FETCh:VOLTage:MAXimum?

Keysight 66312A, 66332A Only

These queries return the maximum output voltage reading from the measurement sample.

Query Syntax	MEASure[:SCALar]:VOLTage:MAXimum? FETCh[:SCALar]:VOLTage:MAXimum?	
Parameters	None	
Examples	MEAS:VOLT:MAX?	FETC:VOLT:MAX?
Returned Parameters	<NR3>	
Related Commands	MEAS:VOLT:MIN?	

MEASure:VOLTage:MINimum? FETCh:VOLTage:MINimum?

Keysight 66312A, 66332A Only

These queries return the minimum output voltage reading from the measurement sample.

Query Syntax	MEASure[:SCALar]:VOLTage:MINimum? FETCh[:SCALar]:VOLTage:MINimum?	
Parameters	None	
Examples	MEAS:VOLT:MIN?	FETC:VOLT:MIN?
Returned Parameters	<NR3>	
Related Commands	MEAS:VOLT:MAX?	

SENSe:CURRent:RANGe

This command selects the dc current measurement range. All models have two current measurement ranges:

High Range: 0 through MAX (see Table 4-3)

Low Range: 0 through 0.02 A (all models)

The High range covers the full current measurement capability of the instrument. The Low range measures currents up to a maximum of 20 mA. This increases the low current measurement sensitivity for greater accuracy and resolution. The value that you program with SENSe:CURRent:RANGe must be the maximum current that you expect to measure. The instrument will select the range that gives the best resolution. The crossover value is 20 mA. When queried, the returned value is the maximum current that can be measured on the range that is presently set.

Command Syntax	SENSe:CURRent[:DC]:RANGe[:UPPer]<NRf+>
Parameters	0 through MAX (see table 4-3)
Unit	A (amperes)
*RST Value	MAX (high range)
Examples	SENS:CURR:RANG 4.0
Query Syntax	SENSe:CURRent:RANGe?
Returned Parameters	<NR3>

SENSe:CURRent:DETEctor**Keysight 66312A, 66332A Only**

This command lets you select the type of detector used for output current measurements. Two choices for detecting current measurements are available:

ACDC This is the preferred choice for all dynamic current measurements. When ACDC is selected, the measured output current includes the current that flows in the instrument's output capacitor. It is especially important to use ACDC detection when measuring pulse or other waveforms with frequency contents greater than several kilohertz.

DC Select DC only if you are making dc current measurements and you require a dc measurement offset accuracy better than 2mA on the High current measurement range. When DC is selected, the component of output current that is supplied by the instrument's output filter is not sensed. Note that this selection gives inaccurate results on current waveforms with frequency contents greater than several kilohertz.

NOTE: This command only applies to the High current measurement range.

Command Syntax	SENSe:CURRent:DETEctor<detector>
Parameters	ACDC or DC
*RST Value	ACDC
Examples	SENS:CURR:DET ACDC SENS:CURR:DET DC
Query Syntax	SENSe:CURRent:DETEct?
Returned Parameters	<CRD>

SENSe:FUNCTION**Keysight 66312A, 66332A Only**

This command configures the measurement sensor to measure either voltage or current when an acquire trigger is used. The query returns the function setting, either VOLT or CURR.

Command Syntax	SENSe:FUNCTION <function
Parameters	"VOLTage" "CURRrent"
Examples	SENS:FUNC "VOLT"
Query Syntax	SENSe:FUNCTION?
Returned Parameters	<SRD>

SENSe:SWEep:OFFSet:POINts**Keysight 66312A, 66332A Only**

This command defines the offset in a data sweep when an acquire trigger is used. Negative values represent data samples taken prior to the trigger.

Command Syntax	SENSe:SWEep:OFFSet:POINts <NRf+>
Parameters	-4095 through 2,000,000,000
*RST Value	0
Examples	SENS:SWE:OFFS:POIN -2047
Query Syntax	SENSe:SWEep:OFFSet:POINts?
Returned Parameters	<NR3>
Related Commands	SENS:SWE:TINT SENS:SWE:POIN MEAS:ARR

SENSe:SWEep:POINts

This command defines the number of points in a data sweep.

Command Syntax	SENSe:SWEep:POINts<NRf+>
Parameters	0 through 4096
*RST Value	2048
Examples	SENS:SWE:POIN 1024
Query Syntax	SENSe:SWEep:POINts?
Returned Parameters	<NR3>
Related Commands	SENS:SWE:TINT SENS:SWE:OFFS MEAS:ARR

SENSe:SWEep:TINTerval

This command defines the time period between samples. The value that you enter for the time interval will be rounded to the nearest 15.6 microsecond increment.

Command Syntax	SENSe:SWEep:TINTerval<NRf+>
Parameters	15.6 microseconds through 31200 seconds
*RST Value	15.6 microseconds
Examples	SENS:SWE:TINT 31.2E-6
Query Syntax	SENSe:SWEep:TINTerval?
Returned Parameters	<NR3>
Related Commands	SENS:SWE:POIN SENS:SWE:OFFS MEAS:ARR

SENSe:WINDow

This command sets the window function that is used in output measurement calculations. The following functions can be selected:

HANNing	A signal conditioning window that reduces errors in dc and rms measurement calculations in the presence of periodic signals such as line ripple. It also reduces jitter when measuring successive pulse waveforms. The Hanning window multiplies each point in the measurement sample by the function $\cos^4$. Do not use the Hanning window when measuring single-shot pulse waveforms.
RECTangular	A window that returns measurement calculations without any signal conditioning. This window may be used for pulse measurements where the exact period of the pulse waveform is known and the measurement interval can be set accordingly using the SENSE:SWEep:TINTerval command.

NOTE: Neither window function alters the instantaneous voltage or current data returned in the measurement array.

Command Syntax	SENSe:WINDow[:TYPE] <type>
Parameters	HANNing RECTangular
*RST Value	HANNing
Examples	SENS:WIND RECT
Query Syntax	SENSe:WINDow[:TYPE]?
Returned Parameters	<CRD>

Output Commands

Output commands consist of output and source commands.

Output commands control the output and digital port functions. They also control the output relay on units with Relay Option 760.

Source commands program the actual voltage, current, and digital port output.

OUTPut

This command enables or disables the dc source output. The state of a disabled output is a condition of zero output voltage and a model-dependent minimum source current (see ***RST**). Unless the NORelay command is programmed, the OUTPut command also controls the output relay on Keysight models 66332A, 6631B, 6632B, 6633B, and 6634B with Relay Option 760. If the NORelay command is sent, the output relay state does NOT change.

Command Syntax	OUTPut[:STATe] <bool> [,NORelay]
Parameters	0 OFF 1 ON
*RST Value	0
Examples	OUTP 1 OUTPUT:STATE ON
Query Syntax	OUTPut[:STATe]?
Returned Parameters	<NR1>0 or 1
Related Commands	*RST *RCL *SAV

OUTPut:DFI

This command enables or disables the discrete fault indicator (DFI) output from the dc source.

Command Syntax	OUTPut:DFI[:STATe]<bool>
Parameters	0 1 OFF ON
*RST Value	OFF
Examples	OUTP:DFI 1 OUTP:DFI ON
Query Syntax	OUTPut:DFI[:STATe]?
Returned Parameters	0 1
Related Commands	OUTP:DFI:SOUR

OUTPut:DFI:SOURce

This command selects the source for discrete fault indicator (DFI) events. The choices are:

QUESTionable	selects the Questionable event summary bit (bit 3 of the Status Byte Register)
OPERation	selects the Operation Event summary bit (bit 7 of the Status Byte Register)
ESB	selects the Standard Event summary bit (bit 5 of the Status Byte Register)
RQS	selects the Request Service bit (bit 6 of the Status Byte Register)
OFF	selects no DFI source

Command Syntax	OUTP:DFI:SOUR<source>
Parameters	QUES OPER ESB RQS OFF
*RST Value	OFF
Examples	OUTP:DFI:SOUR OPER
Query Syntax	OUTPut:DFI:SOUR?
Returned Parameters	<CRD>
Related Commands	OUTP:DFI

OUTPut:PON:STATe

This command selects the power-on state of the dc source. This information is saved in non-volatile memory. The following states can be selected:

- RST** Sets the power-on state to *RST. Refer to the *RST command as described in this chapter for more information.
- RCL0** Sets the power-on state to *RCL 0. Refer to the *RCL command as described in this chapter for more information.

Command Syntax	OUTPut:PON:STATe <state>
Parameters	RST RCL0
Examples	OUTP:PON:STAT RST
Query Syntax	OUTPut:PON:STATe?
Returned Parameters	<CRD>
Related Commands	*RST *RCL

OUTPut:PROTection:CLEAr

This command clears the latch that disables the output when an OverVoltage, OverCurrent, OverTemperature, Remote Inhibit, or Fuse Status condition is detected. All conditions that generate the fault must be removed before the latch can be cleared. The output is then restored to the state it was in before the fault condition occurred.

Command Syntax	OUTPut:PROTection:CLEAr
Parameters	None
Examples	OUTP:PROT:CLE
Related Commands	OUTP:PROT:DEL *RCL *SAV

OUTPut:PROTection:DELAy

This command sets the time between the programming of an output change that produces a constant current condition (CC) and the recording of that condition by the Operation Status Condition register. The delay prevents the momentary changes in status that can occur during reprogramming from being registered as events by the status subsystem. Since the constant current condition is used to trigger overcurrent protection (OCP), this command also delays OCP. Overvoltage protection is not affected by this command.

Command Syntax	OUTPut:PROTection:DELAy <NRf+>
Parameters	0 to 2,147,483.647
Unit	seconds
*RST Value	0.08 (Normal)
Examples	OUTPUT:PROTECTION:DELAY 75E-1
Query Syntax	OUTPut:PROTection:DELAy?
Returned Parameters	<NR3>
Related Commands	OUTP:PROT:CLE *RCL *SAV

OUTPut:RELAy

Keysight 66332A, 6632B, 6633B, 6634B, 6611C, 6612C, 6613C, 6614C Only

This command is only valid for units with Relay Option 760, otherwise an error will occur. Programming ON closes the output relay contacts; programming OFF opens them. The relay is controlled independently of the output state. If the dc source is supplying power to a load, that power will appear at the relay contacts during switching.

Command Syntax	OUTPut:RELAy[:STATe]<bool>
Parameters	0 1 OFF ON
*RST Value	0
Examples	OUTP:REL 1 OUTP:REL OFF
Query Syntax	OUTPput:RELAy?
Returned Parameters	0 1
Related Commands	OUTP *RCL *SAV

OUTPut:RELAy:POLarity

Keysight 66332A, 6632B, 6633B, 6634B, 6611C, 6612C, 6613C, 6614C Only

This command is only valid for units with Relay Option 760, otherwise an error will occur. Programming NORMAl causes the output relay polarity to be the same as the dc source output. Programming REVerse causes the relay output polarity to be opposite to that of the dc source output. If OUTPut = ON when either command is sent, the output voltage is set to 0 during the time that the relays are changing polarity.

Command Syntax	OUTPut:RELAy:POLarity<CRD>
Parameters	NORMAl REVerse
*RST Value	NORM
Examples	OUTP:REL:POL NORM
Query Syntax	OUTPput:RELAy:POLarity?
Returned Parameters	NORM REV
Related Commands	OUTP *RCL *SAV

OUTPut:RI:MODE

This command selects the mode of operation of the Remote Inhibit protection. The RI mode is stored in non-volatile memory. The following modes can be selected:

LATChing	causes a TTL low signal on the INH input to disable the output. The only way to clear the latch is by sending an OUTPut:PROTection:CLEAR command while the INH input is false.
LIVE	allows the INH input to disable the output in a non-latching manner. In other words, the output follows the state of the INH input. When INH is low true, the output is disabled. When INH is high the output is not affected.
OFF	the INH input is disabled.

Command Syntax	OUTPut:RI:MODE <mode>
Parameters	LATChing LIVE OFF
Examples	OUTP:RI:MODE LIVE
Query Syntax	OUTPut:RI:MODE?
Returned Parameters	<CRD>
Related Commands	OUTP:PROT:CLE

[SOURce:]CURRent

This command sets the immediate current level of the dc source . The immediate level is the current programmed for the output terminals.

Command Syntax	[SOURce]:CURRent[:LEVel][:IMMediate][:AMPLitude]<NRf+>
Parameters	see Table 4-3
Default Suffix	A (amperes)
*RST Value	10% of MAX
Examples	CURR 200 MA CURRENT:LEVEL 200 MA
Query Syntax	[SOURce]:CURRent[:LEVel][:IMMediate][:AMPLitude]?
Returned Parameters	<NR3>
Related Commands	CURR:TRIG

[SOURce:]CURRent:TRIGger

This command sets the pending triggered current level of the dc source . The pending triggered level is a stored current value that is transferred to the output terminals when a trigger occurs. In order for a trigger to occur, the trigger subsystem must be initiated (see the INITiate command in the trigger subsystem).

Command Syntax	[SOURce]:CURRent[:LEVel]:TRIGgered[:AMPLitude]<NRf+>
Parameters	see Table 4-3
Default Suffix	A (amperes)
*RST Value	10% of MAX
Examples	CURR:TRIG 1 CURRENT:LEVEL:TRIGGERED 1
Query Syntax	SOURce]:CURRent[:LEVel]:TRIGgered[:AMPLitude]?
Returned Parameters	<NR3>
Related Commands	INIT CURR

[SOURce:]CURRent:PROTection:STATe

This command enables or disables the overcurrent protection (OCP) function. If the dc source overcurrent protection function is enabled and the dc source goes into constant current operation, then the output is disabled and the Questionable Condition status register OC bit is set (see chapter 3 under Programming the Status Registers). Note that the [SOURce:]CURRent command sets the current limit, which determines when the dc source goes into constant current operation. An overcurrent condition can be cleared with the OUTPut:PROTection:CLEar command after the cause of the condition is removed.

NOTE: Use OUTP:PROT:DEL to prevent momentary current limit conditions caused by programmed output changes from tripping the overcurrent protection.

Command Syntax	[SOURce]:CURRent:PROTection:STATe <bool>
Parameters	0 1 OFF ON
*RST Value	OFF
Examples	CURR:PROT:STAT 0 CURRENT:PROTECTION:STATE OFF CURR:PROT:STAT 1 CURRENT:PROTECTION:STATE ON
Query Syntax	Syntax [SOURce]:CURRent:PROTection:STATe?
Returned Parameters	<NR1>0 or 1
Related Commands	OUTP:PROT:CLE *RST

[SOURce:]DIGital:DATA

This command sets and reads the dc source digital control port when that port is configured for Digital I/O operation. The port has three signal pins and a digital ground pin. Pins 1 and 2 are output pins controlled by bits 0 and 1. Pin 3 is controlled by bit 2, and can be programmed to serve either as an input or an output. It normally serves as an output. Bit 2 must be programmed high to use pin 3 as an input. Pin 4 is the digital ground. The query returns the last programmed value in bits 0 and 1 and the value read at pin 3 in bit 2.

Program Value	Bit Configuration			Pin Setting			
	2	1	0	4	3	2	1
0	0	0	0	GND	Input	Lo	Lo
1	0	0	1	GND	Input	Lo	Hi
2	0	1	0	GND	Input	Hi	Lo
3	0	1	1	GND	Input	Hi	Hi
4	1	0	0	GND	Output	Lo	Lo
5	1	0	1	GND	Output	Lo	Hi
6	1	1	0	GND	Output	Hi	Lo
7	1	1	1	GND	Output	Hi	Hi

Command Syntax [SOURce]:DIGital:DATA[:VALue] <NRf>
Parameters 0 to 7
***RST Value** 0
Examples DIG:DATA 7
Query Syntax [SOURce]:DIGital:DATA?
Returned Parameters <NR1>
Related Commands DIG:FUNC

[SOURce:]DIGital:FUNCTION

This command configures the dc source digital control port. The configuration setting is saved in non-volatile memory.

RIDFi Configures the port for Remote Inhibit/Discrete Fault Interrupt operation
DIGio Configures the port for Digital input/output operation (see DIG:DATA)

Command Syntax [SOURce]:DIGital:FUNCTION <CRD>
Parameters RIDFi | DIGio
Examples DIG:FUNC DIG
Query Syntax [SOURce]:DIGital:FUNC?
Returned Parameters <CRD>
Related Commands DIG:DATA

[SOURce:]VOLTage

This command sets the output voltage level of the dc source.

Command Syntax [SOURce]:VOLTage[:LEVel][:IMMediate][:AMPLitude]<NRf+>
Parameters see Table 4-3
Default Suffix V (volts)
***RST Value** 0
Examples VOLT 2 VOLTAGE:LEVEL 200 MV
Query Syntax [SOURce]:VOLTage[:LEVel][:IMMediate][:AMPLitude]?
Returned Parameters <NR3>
Related Commands VOLT:TRIG

[SOURce:]VOLTage:ALC:BANDwidth?**[SOURce:]VOLTage:ALC:BWIDth?****Keysight 66332A, 6631B, 6632B, 6633B and 6634B Only**

These queries return the setting of the output mode switch. The output mode switch is used to connect or disconnect the the output capacitor located inside the unit. The returned value is 15,000 if the switch is set to Normal and 60,000 if the switch is set to Fast.

Query Syntax	[SOURce]:VOLTage:ALC:BANDwidth?
	[SOURce]:VOLTage:ALC:BWIDth?
Examples	VOLT:ALC:BAND? VOLTAGE:ALC:BWIDth?
Returned Parameters	<NR3>

[SOURce:]VOLTage:TRIGger

This command sets the pending triggered voltage level of the dc source. The pending triggered level is a stored voltage value that is transferred to the output terminals when a trigger occurs. In order for a trigger to occur, the trigger subsystem must be initiated (see the INITiate command in the trigger subsystem).

Command Syntax	[SOURce][:VOLTage[:LEVel]:TRIGgered[:AMPLitude]<NRf+>
Parameters	see Table 4-3
Default Suffix	V (volts)
*RST Value	0
Examples	VOLT:TRIG 20 VOLTAGE:LEVEL:TRIGGERED 20
Query Syntax	[SOURce]:VOLTage[LEVel]:TRIGgered[:AMPLitude]?
Returned Parameters	<NR3>
Related Commands	VOLT *RST

[SOURce:]VOLTage:PROTection

This command sets the overvoltage protection (OVP) level of the dc source. If the output voltage exceeds the OVP level, then the dc source output is disabled and the Questionable Condition status register OV bit is set (see chapter 3 under Programming the Status Registers). An overvoltage condition can be cleared with the OUTP:PROT:CLE command after the condition that caused the OVP trip is removed. The OVP always trips with zero delay and is unaffected by the OUTP:PROT:DEL command.

Command Syntax	[SOURce]:VOLTage:PROTection[:LEVel]<NRf+>
Parameters	see Table 4-3
Default Suffix	V (volts)
*RST Value	MAX
Examples	VOLT:PROT 21.5 VOLT:PROT:LEV MAX
Query Syntax	[SOURce]:VOLTage:PROTection[:LEVel]?
Returned Parameters	<NR3>
Related Commands	OUTP:PROT:CLE OUTP:PROT:DEL

Status Commands

Status commands program the dc source status registers. The dc source has three groups of status registers; Operation, Questionable, and Standard Event. The Standard Event group is programmed with Common commands as described later in this section. The Operation and Questionable status groups each consist of the Condition, Enable, and Event registers and the NTR and PTR filters. Chapter 3 under "Programming the Status Registers" explains how to read specific register bits and use the information they return.

Common commands also perform status functions. The following common commands are discussed in this section: *CLS *ESE *SR? *OPC *PSC *SRE *STB *WAI.

STATus:PRESet

This command sets all defined bits in the Status Subsystem PTR registers and clears all bits in the subsystem NTR and Enable registers.

Command Syntax STATus:PRESet
Parameters None
Examples STAT:PRESet STATUS:PRESET

Table 4-4. Bit Configuration of Operation Status Registers

Bit Position	15–12	11	10	9	8	7-6	5	4-1	0
Bit Name	not used	CC-	CC+	not used	CV	not used	WTG	not used	CAL
Bit Weight		2048	1024		256		32		1
CAL = The dc source is computing new calibration constants. WTG = The dc source is waiting for a trigger. CV = The dc source is operating in constant voltage mode. CC+ = The dc source is operating in constant current mode. CC = The dc source is operating in negative constant current mode.									

STATus:OPERation?

This query returns the value of the Operation Event register. The Event register is a read-only register which holds (latches) all events that are passed by the Operation NTR and/or PTR filter. Reading the Operation Event register clears it.

Query Syntax STATus:OPERtion[:EVENT]?
Parameters None
Returned Parameters <NR1>(Register Value)
Examples STAT:OPER? STATUS:OPERATIONAL:EVENT?
Related Commands *CLS STAT:OPER:NTR STAT:OPER:PTR

STATus:OPERation:CONDition?

This query returns the value of the Operation Condition register. That is a read-only register which holds the real-time (unlatched) operational status of the dc source .

Query Syntax STATus:OPERation:CONDition?
Parameters None
Examples STAT:OPER:COND? STATUS:OPERATION:CONDITION?
Returned Parameters <NR1> (register value)

STATus:OPERation:ENABLE

This command and its query set and read the value of the Operational Enable register. This register is a mask for enabling specific bits from the Operation Event register to set the operation summary bit (OPER) of the Status Byte register. This bit (bit 7) is the logical OR of all the Operational Event register bits that are enabled by the Status Operation Enable register.

Command Syntax	STATus:OPERation:ENABLE<NRf>
Parameters	0 to 32727
Preset Value	0
Examples	STAT:OPER:ENAB 1312 STAT:OPER:ENAB 1 STATUS:OPERATION:ENABLE?
Query Syntax	STATus:OPERation:ENABLE?
Returned Parameters	<NR1> (register value)
Related Commands	STAT:OPER:EVEN

STATus:OPERation:NTR**STATus:OPERation:PTR**

These commands set or read the value of the Operation NTR (Negative-Transition) and PTR (Positive-Transition) registers. These registers serve as polarity filters between the Operation Enable and Operation Event registers to cause the following actions:

- ◆ When a bit in the Operation NTR register is set to 1, then a 1-to-0 transition of the corresponding bit in the Operation Condition register causes that bit in the Operation Event register to be set.
- ◆ When a bit of the Operation PTR register is set to 1, then a 0-to-1 transition of the corresponding bit in the Operation Condition register causes that bit in the Operation Event register to be set.
- ◆ If the same bits in both NTR and PTR registers are set to 1, then any transition of that bit at the Operation Condition register sets the corresponding bit in the Operation Event register.
- ◆ If the same bits in both NTR and PTR registers are set to 0, then no transition of that bit at the Operation Condition register can set the corresponding bit in the Operation Event register.

Command Syntax	STATus:OPERtion:NTRansition<NRf> STATus:OPERtion:PTRansition<NRf>
Parameters	0 to 32727
Preset Value	NTR register = 0; PTR register = 32727
Examples	STAT:OPER:NTR 32 STAT:OPER:PTR 1312
Query Syntax	STAT:OPER:NTR? STAT:OPER:PTR?
Returned Parameters	<NR1> (register value)
Related Commands	STAT:OPER:ENAB

Table 4-5. Bit Configuration of Questionable Status Registers

Bit Position	15	14	13-11	10	9	8-5	4	3	2	1	0
Bit Name	not used	Meas Ovld	not used	Unreg	RI	not used	OT	not used	FS	OCP	OV
Bit Weight		16384		1024	512		16		4	2	1
OV = overvoltage protection has tripped OCP = overcurrent protection has tripped FS = the fuse is blown OT = overtemperature protection has tripped RI = remote inhibit is active Unreg = output is unregulated Meas Ovld = measurement overload											

STATus:QUEStionable?

This query returns the value of the Questionable Event register. The Event register is a read-only register which holds (latches) all events that are passed by the Questionable NTR and/or PTR filter. Reading the Questionable Event register clears it.

Query Syntax STATus:QUEStionable[:EVENT]?
Parameters None
Examples STAT:QUES? STATUS:QUESTIONABLE:EVENT?
Returned Parameters <NR1> (register value)
Related Commands *CLS STAT:QUES:ENAB
 STAT:QUES:NTR STAT:QUES:PTR

STATus:QUEStionable:CONDition?

This query returns the value of the Questionable Condition register. That is a read-only register which holds the real-time (unlatched) questionable status of the dc source.

Query Syntax STATus:QUEStionable:CONDition?
Parameters None
Examples STAT:QUES:COND? STATUS:QUESTIONABLE:CONDITION?
Returned Parameters <NR1> (register value)

STATus:QUEStionable:ENABLE

This command and its query set and read the value of the Questionable Enable register. This register is a mask for enabling specific bits from the Questionable Event register to set the questionable summary bit (QUES) of the Status Byte register. This bit (bit 3) is the logical OR of all the Questionable Event register bits that are enabled by the Questionable Status Enable register..

Command Syntax STATus:QUEStionable:ENABLE<NRf>
Parameters 0 to 32767
Preset Value 0
Examples STAT:QUES:ENAB 20 STAT:QUES:ENAB 16
Query Syntax STATus:QUEStionable:ENABLE?
Returned Parameters <NR1> (register value)
Related Commands STAT:QUES?

STATus:QUEStionable:NTR

STATus:QUEStionable:PTR

These commands allow you to set or read the value of the Questionable NTR (Negative-Transition) and PTR (Positive-Transition) registers. These registers serve as polarity filters between the Questionable Enable and Questionable Event registers to cause the following actions:

- ◆ When a bit of the Questionable NTR register is set to 1, then a 1-to-0 transition of the corresponding bit of the Questionable Condition register causes that bit in the Questionable Event register to be set.
- ◆ When a bit of the Questionable PTR register is set to 1, then a 0-to-1 transition of the corresponding bit in the Questionable Condition register causes that bit in the Questionable Event register to be set.
- ◆ If the same bits in both NTR and PTR registers are set to 1, then any transition of that bit at the Questionable Condition register sets the corresponding bit in the Questionable Event register.
- ◆ If the same bits in both NTR and PTR registers are set to 0, then no transition of that bit at the Questionable Condition register can set the corresponding bit in the Questionable Event register.

Command Syntax	STATus:QUEStionable:NTRansition<NRf> STATus:QUEStionable:PTRansition<NRf>
Parameters	0 to 32727
Preset Value	NTR register = 0; PTR register = 32727
Examples	STAT:QUES:NTR 16 STATUS:QUESTIONABLE:PTR 512
Query Syntax	STAT:QUES:NTR?STAT:QUES:PTR?
Returned Parameters	<NR1>(Register value)
Related Commands	STAT:QUES:ENAB

*CLS

This command causes the following actions (see chapter 3 under Programming the Status Registers, for the descriptions of all registers):

- ◆ Clears the following registers:
 - Standard Event Status
 - Operation Status Event
 - Questionable Status Event
 - Status Byte
- ◆ Clears the Error Queue
- ◆ If *CLS immediately follows a program message terminator (<NL>), then the output queue and the MAV bit are also cleared.

Command Syntax	*CLS
Parameters	None

***ESE**

This command programs the Standard Event Status Enable register bits. The programming determines which events of the Standard Event Status Event register (see *ESR?) are allowed to set the ESB (Event Summary Bit) of the Status Byte register. A "1" in the bit position enables the corresponding event. All of the enabled events of the Standard Event Status Event Register are logically ORed to cause the Event Summary Bit (ESB) of the Status Byte Register to be set. The query reads the Standard Event The query reads the Standard Event Status Enable register.

Table 4-6. Bit Configuration of Standard Event Status Enable Register

Bit Position	7	6	5	4	3	2	1	0
Bit Name	PON	0	CME	EXE	DDE	QUE	0	OPC
Bit Weight	128	64	32	16	8	4	2	1
PON = Power-on has occurred CME = Command error EXE = Execution error				DDE = Device-dependent error QUE = Query error OPC = Operation complete				

Command Syntax *ESE <NRf>
Parameters 0 to 255
Power-On Value (See *PSC)
Examples *ESE 129
Query Syntax *ESE?
Returned Parameters <NR1>(Register value)
Related Commands *ESR? *PSC *STB?

CAUTION: If *PSC is programmed to 0, the *ESE command causes a write cycle to nonvolatile memory. Nonvolatile memory has a finite maximum number of write cycles. Programs that repeatedly cause write cycles to nonvolatile memory can eventually exceed the maximum number of write cycles and cause the memory to fail.

***ESR?**

This query reads the Standard Event Status Event register. Reading the register clears it. The bit configuration is the same as the Standard Event Status Enable register (see *ESE).

Query Syntax *ESR?
Parameters None
Returned Parameters <NR1>(Register binary value)
Related Commands *CLS *ESE *ESE? *OPC

***OPC**

This command causes the instrument to set the OPC bit (bit 0) of the Standard Event Status register when the has completed all pending operations. (See *ESE for the bit configuration of the Standard Event Status register.) *Pending operations* are complete when:

- ◆ all commands sent before *OPC have been executed. This includes overlapped commands. Most commands are sequential and are completed before the next command is executed. Overlapped commands are executed in parallel with other commands. Commands that affect output voltage, current or state, relays, and trigger actions are overlapped with subsequent commands sent to the dc source. The *OPC command provides notification that all overlapped commands have been completed.
- ◆ all triggered actions are completed

4 - Language Dictionary

* OPC does not prevent processing of subsequent commands, but bit 0 will not be set until all pending operations are completed.

*OPC? causes the instrument to place an ASCII "1" in the Output Queue when all pending operations are completed. Unlike *OPC, *OPC? prevents processing of all subsequent commands. It is intended to be used at the end of a command line so that the application program can then monitor the bus for data until it receives the "1" from the dc source Output Queue.

Command Syntax	*OPC
Parameters	None
Query Syntax	*OPC?
Returned Parameters	<NR1> 1
Related Commands	*OPC *TRIG *WAI

*PSC

This command controls the automatic clearing at power-on of the Service Request Enable and the Standard Event Status Enable registers

***PSC ON | 1** causes these registers to be cleared at power-on. This prevents a PON event from generating SRQ at power-on.

***PSC OFF | 0** causes the contents of the Standard Event Enable and Service Request Enable registers to be saved in nonvolatile memory and recalled at power-on. This allows a PON event to generate SRQ at power-on.

The query returns the current state of *PSC.

Command Syntax	*PSC <Bool>
Parameters	0 1 OFF ON
Example	*PSC 0 *PSC 1
Query Syntax	*PSC?
Returned Parameters	<NR1>0 1
Related Commands	*ESE *SRE

CAUTION: *PSC causes a write cycle to nonvolatile memory. Nonvolatile memory has a finite maximum number of write cycles. Programs that repeatedly cause write cycles to nonvolatile memory can eventually exceed the maximum number of write cycles and cause the memory to fail.

*SRE

This command sets the condition of the Service Request Enable Register. This register determines which bits from the Status Byte Register (see *STB for its bit configuration) are allowed to set the Master Status Summary (MSS) bit and the Request for Service (RQS) summary bit. A 1 in any Service Request Enable Register bit position enables the corresponding Status Byte Register bit and all such enabled bits then are logically ORed to cause Bit 6 of the Status Byte Register to be set.

When the controller conducts a serial poll in response to SRQ, the RQS bit is cleared, but the MSS bit is not. When *SRE is cleared (by programming it with 0), the dc source cannot generate an SRQ to the controller.

The query returns the current state of *SRE.

Command Syntax	*SRE <NRf>
Parameters	0 to 255
Power-on Value	see *PSC
Example	*SRE 20
Query Syntax	*SRE?
Returned Parameters	<NR1> (register binary value)
Related Commands	*ESE *ESR *PSC

CAUTION: If *PSC is programmed to 0, the *SRE command causes a write cycle to nonvolatile memory. Nonvolatile memory has a finite maximum number of write cycles. Programs that repeatedly cause write cycles to nonvolatile memory can eventually exceed the maximum number of write cycles and cause the memory to fail.

*STB?

This query reads the Status Byte register, which contains the status summary bits and the Output Queue MAV bit. Reading the Status Byte register does not clear it. The input summary bits are cleared when the appropriate event registers are read. The MAV bit is cleared at power-on, by *CLS' or when there is no more response data available.

A serial poll also returns the value of the Status Byte register, except that bit 6 returns Request for Service (RQS) instead of Master Status Summary (MSS). A serial poll clears RQS, but not MSS. When MSS is set, it indicates that the has one or more reasons for requesting service.

Table 4-7. Bit Configuration of Status Byte Register

Bit Position	7	6	5	4	3	2	1	0
Bit Name	OPER	MSS (RQS)	ESB	MAV	QUES	0	0	0
Bit Weight	128	64	32	16	8	4	2	1
ESB = Event status byte summary MAV = Message available MSS = Master status summary				OPER = Operation status summary QUES = Questionable status summary RQS = Request for service				

Query Syntax	*STB?
Returned Parameters	<NR1>(Register binary value)

*WAI

This command instructs the dc source not to process any further commands until all pending operations are completed. "Pending operations" are as defined under the *OPC command. *WAI can be aborted only by sending the dc source an GPIB DCL (Device Clear) command.

Command Syntax	WAI?
Parameters	None
Related Commands	*OPC*OPC?

System Commands

System commands consist of system, display, and common commands.

System commands commands control system functions that are not directly related to output control or measurement functions.

Display commands control the front panel display of the .

Common commands also perform system functions. The following common commands are discussed in this section: *IDN? *OPT? *RCL *RST *SAV *TST?.

DISPlay

This command turns the front panel display on or off. When off, the front panel display is blank. The display annunciators are not affected by this command.

Command Syntax	DISPlay[:WINDow][:STATe] <bool>
Parameters	0 1 OFF ON
*RST Value	ON
Examples	DISP ON DISPLAY:STATE ON
Query Syntax	DISPlay[:WINDow][STATe]?
Returned Parameters	<NR1> 0 or 1
Related Commands	DISP:MODE DISP:TEXT *RST

DISPlay:MODE

Switches the display between its normal instrument functions and a mode in which it displays text sent by the user. Text messages are defined with the DISPlay:TEXT command.

Command Syntax	DISPlay[:WINDow]:MODE NORMa TEXT
Parameters	<CRD>NORMa TEXT
*RST Value	NORM
Examples	DISP:MODE NORM DISPLAY:MODE TEXT
Query Syntax	DISPlay[:WINDow]:MODE?
Returned Parameters	<CRD> NORMAL or TEXT
Related Commands	DISP DISP:TEXT *RST

DISPlay:TEXT

This command sends character strings to the display when the display mode is set to TEXT. The character string is case-sensitive and must be enclosed in either single (') or double (") quotes. The display is capable of showing up to 14 characters. Strings exceeding 14 characters will be truncated.

Command Syntax	DISPlay[:WINDow]:TEXT [:DATA] <display_string>
Parameters	<display string>
*RST Value	null string
Examples	DISP:TEXT "DEFAULT_MODE" DISPLAY:WINDOW:TEXT:DATA `533.2E-1VOLTS`
Query Syntax	DISPlay[:WINDow]:TEXT?
Returned Parameters	<STR>(Last programmed text string)
Related Commands	DISP DISP:MODE

SYSTem:ERRor?

This query returns the next error number followed by its corresponding error message string from the remote programming error queue. The queue is a FIFO (first-in, first-out) buffer that stores errors as they occur. As it is read, each error is removed from the queue. When all errors have been read, the query returns 0,NO ERROR. If more errors are accumulated than the queue can hold, the last error in the queue will be -350,TOO MANY ERRORS (see Appendix C for other error codes).

You can use the front panel Error key to read errors from the queue. Errors generated at the front panel are not put into the queue but appear immediately on the display.

Query Syntax	SYSTem:ERRor?
Parameters	(None)
Returned Parameters	<NR1>, <SRD>
Examples	SYST:ERR? SYSTEM:ERROR?

SYSTem:LANGUage

This command switches the instrument between its SCPI command language and its compatibility language. The compatibility language is provided for emulation of older dc source systems and is described in Appendix B . Sending the command causes:

- ◆ The selected language to become active and to be stored in nonvolatile memory.
- ◆ The to reset to its power-on state.

If the dc source is shut off, it will resume operation in the last-selected language when power is restored. Note that this command and query can be used regardless of the language that is presently selected.

Command Syntax	SYSTem:LANGUage<string>
Parameters	SCPI COMPatibility
Power-on Value	<i>last selected language</i>
Example	SYST:LANG SCPI SYSTEM:LANGUAGE COMPATIBILITY
Query Syntax	SYSTem:LANGUage?
Returned Parameters	<CRD>

SYSTem:VERSion?

This query returns the SCPI version number to which the complies. The returned value is of the form YYYY.V, where YYYY represents the year and V is the revision number for that year.

Query Syntax	SYSTem:VERSion?
Parameters	(none)
Returned Parameters	<NR2>
Examples	SYST:VERS? SYSTEM:VERSION?

SYSTem:LOCal

For RS-232 Operation Only

This command places the dc source in local mode during RS-232 operation. The front panel keys are functional.

Command Syntax	SYSTem:LOCal
Parameters	None
Example	SYST:LOC
Related Commands	SYST:REM SYST:RWL

SYSTem:REMOte

For RS-232 Operation Only

This command places the dc source in remote mode during RS-232 operation. This disables all front panel keys except the Local key. Pressing the Local key while in the remote state returns the front panel to the local state.

Command Syntax	SYSTem:REMOte
Parameters	None
Example	SYST:REM
Related Commands	SYST:LOC SYST:RWL

SYSTem:RWLock

For RS-232 Operation Only

This command places the dc source in remote mode during RS-232 operation. All front panel keys including the Local key are disabled. Use SYSTem:LOCAl to return the front panel to the local state.

Command Syntax	SYSTem:RWLock
Parameters	None
Example	SYST:RWL
Related Commands	SYST:REM SYST:LOC

*IDN?

This query requests the dc source to identify itself. It returns a string composed of four fields separated by commas.

Query Syntax	*IDN?		
Returned Parameters	<AARD>	Field	Information
		Keysight Technologies	Manufacturer
		xxxxxA	model number followed by a letter suffix
		nnnnA-nnnnn	10-character serial number or 0
		<A>.xx.xx	Revision levels of firmware.
Example	KEYSIGHT, 66312A, 0, A.00.01		

***OPT?**

This query requests the dc source to identify any options that are installed. Options are identified by number A 0 indicates no options are installed.

Query Syntax	*OPT?
Returned Parameters	<AARD>

***RCL**

WARNING: Recalling a previously stored state may place hazardous voltages at the dc source output.

This command restores the dc source to a state that was previously stored in memory with the *SAV command to the specified location. All states are recalled with the following exceptions:

- ◆ the trigger system is set to the Idle state by an implied ABORt command (this cancels any uncompleted trigger actions)
- ◆ the calibration function is disabled by setting CAL:STATe to OFF

NOTE: The device state stored in location 0 is automatically recalled at power turn-on when the OUTPut:PON:STATe is set to RCL0.

Command Syntax	*RCL <NRf>
Parameters	0 1 2 3
Example	*RCL 3
Related Commands	*PSC *RST *SAV

***RST**

This command resets the to a factory-defined state as defined in the following table. *RST also forces an ABORt command.

Command Syntax	*RST
Parameters	None
Related Commands	*PSC *SAV

Table 4-8. *RST Settings

CAL:STAT	OFF	[SOUR:]CURR	10% of MAX*
DIG:DATA	0	[SOUR:]CURR:TRIG	10% of MAX*
DISP:STAT	ON	[SOUR:]CURR:PROT:STAT	OFF
DISP:MODE	NORM	[SOUR:]LIST:COUN	0
DISP:TEXT	' '	[SOUR:]VOLT	0
INIT:CONT	OFF	[SOUR:]VOLT:TRIG	0
OUTP	OFF	[SOUR:]VOLT:PROT	MAX*
OUTP:DFI	OFF	TRIG:ACQ:COUN:CURR	1
OUTP:DFI:SOUR	OFF	TRIG:ACQ:COUN:VOLT	1
OUTP:PROT:DEL	.08 Norm; .008 Fast	TRIG:ACQ:HYST:CURR	0
OUTP:REL	OFF	TRIG:ACQ:HYST:VOLT	0
OUTP:REL:POL	NORM	TRIG:ACQ:LEV:CURR	MAX*
SENS:CURR:RANG	MAX	TRIG:ACQ:LEV:VOLT	MAX*
SENS:CURR:DET	ACDC	TRIG:ACQ:SLOP:CURR	POS
SENS:FUNC	VOLT	TRIG:ACQ:SLOP:VOLT	POS
SENS:SWE:OFFS:POIN	0	TRIG:ACQ:SOUR	INTERNAL
SENS:SWE:POIN	2048	TRIG:TRAN:SOUR	BUS
SENS:SWE:TINT	15.6 μ s		

* Maximum values are model-dependent. Refer to Table 4-3.

***SAV**

This command stores the present state of the dc source to the specified location in non-volatile memory. Up to 4 states can be stored. If a particular state is desired at power-on, it should be stored in location 0. It will then be automatically recalled at power turn-on if OUTPut:PON:STATe is set to RCL0. *RCL retrieves instrument states.

Command Syntax *SAV <NRf>
Parameters 0 | 1 | 2 | 3
Example *SAV 3
Related Commands *RCL *RST

CAUTION: *SAV causes a write cycle to nonvolatile memory. Nonvolatile memory has a finite maximum number of write cycles. Programs that repeatedly cause write cycles to nonvolatile memory can eventually exceed the maximum number of write cycles and cause the memory to fail.

***TST?**

This query causes the to do a self-test and report any errors. 0 indicates that the dc source passed self-test. 1 indicates that one or more tests failed. Selftest errors are written to the error queue (see Appendix C).

Query Syntax TST?
Returned Parameters <NR1>

Trigger Commands

Trigger commands consist of trigger and initiate commands.

Trigger commands control the remote triggering of the dc source . Trigger commands (and Initiate commands) are referenced either by name or by number. The correspondence between names and numbers is:

Sequence Number	Sequence Name	Description
1 (the default)	TRANSient	Output transient trigger sequence
2	ACQuire	Measurement acquire trigger sequence

Initiate commands initialize the trigger system.

ABORt

This command cancels any trigger actions presently in process. Pending trigger levels are reset to their corresponding immediate values. ABORt also resets the WTG bit in the Operation Condition Status register (see chapter 3 under Programming the Status Registers). If INITiate:CONTinuous ON has been programmed, the trigger subsystem initiates itself immediately after ABORt, thereby setting WTG. ABORt is executed at power turn on and upon execution of *RCL or RST.

Command Syntax	ABORt
Parameters	None
Examples	ABOR
Related Commands	INIT *RST *TRG TRIG

INITiate:SEQuence

INITiate:NAME

INIT:SEQ2 or INIT:NAME ACQ applies to Keysight 66312A, 66332A Only

INITiate commands control the initiation of both output and measurement triggers. When a trigger is enabled, an event on a selected trigger source causes the specified triggering action to occur. If the trigger subsystem is not enabled, all trigger commands are ignored.

Command Syntax	INITiate[:IMMediate]:SEQuence[1 2] INITiate[:IMMediate]:NAME<name>
Parameters	For INIT:NAME TRANSient ACQuire
Examples	INIT:SEQ2 INIT:NAME TRAN
Related Commands	ABOR INIT:CONT TRIG TRIG:SEQ:DEF *TRG

INITiate:CONTinuous:SEQuence1

INITiate:CONTinuous:NAME

These commands control the output transient trigger system.

1 or ON continuously initiates the output trigger system..
0 or OFF turns off continuous triggering. In this state, the output trigger system must be initiated for each trigger using INITiate:SEQuence.

Command Syntax	INITiate:CONTinuous:SEQuence1<bool> INITiate:CONTinuous:NAME TRANSient,<bool>
Parameters	0 1 OFF ON
Examples	INIT:CONT:SEQ ON INIT:CONT:NAME TRAN, 1
Related Commands	ABOR INIT TRIG TRIG:SEQ:DEF *TRG

TRIGger

When the transient trigger subsystem is initiated, this command generates a trigger signal. The trigger will then:

1. Initiate a pending level change as specified by CURRent:TRIGger or VOLTage:TRIGger.
2. Clear the WTG bit in the Status Operation Condition register after both transient and acquire trigger sequences have completed. (WTG is the logical-or of both transient and acquire sequences.)
3. If INITiate:CONTinuous ON has been programmed, the trigger subsystem is immediately re-enabled for subsequent triggers. As soon as it is cleared, the WTG bit is again set to 1.

Command Syntax	TRIGger[:SEquence1][:IMMediate] TRIGger[:TRANsient][:IMMediate]
Parameters	None
Examples	TRIG TRIG:IMM
Related Commands	ABOR CURR:TRIG INIT *TRG VOLT:TRIG

TRIGger:SOURce

This command is included for completeness. It selects the trigger source for transient triggers. Since BUS is the only trigger source for transient triggers, this command does not need to be used.

BUS GPIB device, *TRG, or <GET> (Group Execute Trigger)

Command Syntax	TRIGger[:SEquence1]:SOURce<source> TRIGger[:TRANsient]:SOURce<source>
Parameters	BUS
*RST Value	BUS
Examples	TRIG:SOUR BUS
Query Syntax	TRIGger[:SEquence1]:SOURce? TRIGger[:TRANsient]:SOURce?
Returned Parameters	<CRD>

TRIGger:SEquence2

TRIGger:ACQuire

Keysight 66312A, 66332A Only

When the trigger subsystem is initiated, these commands generate a measurement trigger signal. The measurement trigger causes the dc source to measure the output voltage and current and store the results in a buffer.

Command Syntax	TRIGger:SEquence2[:IMMediate] TRIGger:ACQuire[:IMMediate]
Parameters	None
Examples	TRIG:SEQ2 TRIG:ACQ
Related Commands	TRIG:SOUR TRIG:SEQ2:DEF TRIG:SEQ2:COUN TRIG:SEQ2:LEV:VOLT TRIG:SEQ2:SLOP:CURRE

TRIGger:SEQuence2:COUNT:CURRent

TRIGger:ACQuire:COUNT:CURRent

Keysight 66312A, 66332A Only

This command sets up a successive number of triggers for measuring current data. With this command, the trigger system needs to be initialized only once at the start of the acquisition period. After each completed measurement, the instrument waits for the next valid trigger condition to start another measurement. This continues until the count has completed.

Command Syntax	TRIGger:SEQuence2:COUNT:CURRent<NRf+> TRIGger:ACQuire:COUNT:CURRent<NRf+>
Parameters	1 to 100
*RST Value	1
Examples	TRIG:SEQ2:COUN:CURR 5 TRIG:ACQ:COUN:CURR 1
Query Syntax	TRIGger:SEQuence2:COUNT:CURRent? TRIGger:ACQuire:COUNT:CURRent?
Returned Parameters	<NR3>
Related Commands	TRIG:SEQ2 TRIG:ACQ

TRIGger:SEQuence2:COUNT:VOLTage

TRIGger:ACQuire:COUNT:VOLTage

Keysight 66312A, 66332A Only

This command sets up a successive number of triggers for measuring voltage data. With this command, the trigger system needs to be initialized only once at the start of the acquisition period. After each completed measurement, the instrument waits for the next valid trigger condition to start another measurement. This continues until the count has completed.

Command Syntax	TRIGger:SEQuence2:COUNT:VOLTage<NRf+> TRIGger:ACQuire:COUNT:VOLTage<NRf+>
Parameters	1 to 100
*RST Value	1
Examples	TRIG:SEQ2:COUN:VOLT 5 TRIG:ACQ:COUN:VOLT 1
Query Syntax	TRIGger:SEQuence2:COUNT:VOLTage? TRIGger:ACQuire:COUNT:VOLTage?
Returned Parameters	<NR3>
Related Commands	TRIG:SEQ2 TRIG:ACQ

TRIGger:SEQuence2:HYSTeresis:CURRent

TRIGger:ACQuire:HYSTeresis:CURRent

Keysight 66312A, 66332A Only

This command defines a band around the trigger level through which the signal must pass before an internal measurement can occur. The band limit above and below the trigger level is one half of the hysteresis value added to or subtracted from the trigger level.

For a positive trigger to occur, the excursion of an output waveform in the positive direction must start below the lower hysteresis band limit and pass through the upper hysteresis band limit. For a negative trigger to occur, the excursion of an output waveform in the negative direction must start above the upper hysteresis band limit and pass through the lower hysteresis band limit.

Command Syntax	TRIGger:SEQuence2:HYSTeresis:CURRent<NRf+> TRIGger:ACQuire:HYSTeresis:CURRent<NRf+>
Parameters	0 to MAX (see table 4-3)
Unit	A (amperes)
*RST Value	0
Examples	TRIG:SEQ2:HYST:CURR 0.5 TRIG:ACQ:HYST:CURR 0.5
Query Syntax	TRIGger:SEQuence2:HYSTeresis:CURRent? TRIGger:ACQuire:HYSTeresis:CURRent?
Returned Parameters	<NR3>
Related Commands	TRIG:SEQ2:HYST:VOLT TRIG:SEQ2:LEV:CURR

TRIGger:SEQuence2:HYSTeresis:VOLTage

TRIGger:ACQuire:HYSTeresis:VOLTage

Keysight 66312A, 66332A Only

This command defines a band around the trigger level through which the signal must pass before an internal measurement can occur. The band limit above and below the trigger level is one half of the hysteresis value added to or subtracted from the trigger level.

For a positive trigger to occur, the excursion of an output waveform in the positive direction must start below the lower hysteresis band limit and pass through the upper hysteresis band limit. For a negative trigger to occur, the excursion of an output waveform in the negative direction must start above the upper hysteresis band limit and pass through the lower hysteresis band limit.

Command Syntax	TRIGger:SEQuence2:HYSTeresis:VOLTage<NRf+> TRIGger:ACQuire:HYSTeresis:VOLTage<NRf+>
Parameters	0 to MAX (see table 4-3)
Unit	V (volts)
*RST Value	0
Examples	TRIG:SEQ2:HYST:VOLT 2 TRIG:ACQ:HYST:VOLT 2
Query Syntax	TRIGger:SEQuence2:HYSTeresis:VOLTage? TRIGger:ACQuire:HYSTeresis:VOLTage?
Returned Parameters	<NR3>
Related Commands	TRIG:SEQ2:HYST:CURR TRIG:SEQ2:LEV:VOLT

TRIGger:SEQuence2:LEVel:CURRent

TRIGger:ACQuire:LEVel:CURRent

Keysight 66312A, 66332A Only

This command sets the trigger level for internally triggered current measurements. A positive current trigger occurs when the current level changes from a value less than the lower hysteresis band limit to a value greater than the upper hysteresis band limit. Similarly, a negative current trigger occurs when the current level changes from a value greater than the upper hysteresis band limit to a value less than the lower hysteresis band limit.

Command Syntax	TRIGger:SEQuence2:LEVel:CURRent<NRf+> TRIGger:ACQuire:LEVel:CURRent<NRf+>
Parameters	0 to MAX (see table 4-3)
Unit	A (amperes)
*RST Value	0
Examples	TRIG:SEQ2:LEV:CURR 5 TRIG:ACQ:LEV:CURR MAX TRIG:ACQ:LEV 2
Query Syntax	TRIGger:SEQuence2:LEVel:CURRent? TRIGger:ACQuire:LEVel:CURRent?
Returned Parameters	<NR3>
Related Commands	TRIG:SEQ2:LEV:VOLT TRIG:SEQ2:HYST:CURR

TRIGger:SEQuence2:LEVel:VOLTage

TRIGger:ACQuire:LEVel:VOLTage

Keysight 66312A, 66332A Only

This command sets the trigger level for internally triggered voltage measurements. A positive voltage trigger occurs when the voltage level changes from a value less than the lower hysteresis band limit to a value greater than the upper hysteresis band limit. Similarly, a negative voltage trigger occurs when the voltage level changes from a value greater than the upper hysteresis band limit to a value less than the lower hysteresis band limit.

Command Syntax	TRIGger:SEQuence2:LEVel:VOLTage<NRf+> TRIGger:ACQuire:LEVel:VOLTage<NRf+>
Parameters	0 to MAX (see table 4-3)
Unit	V (volts)
*RST Value	0
Examples	TRIG:SEQ2:LEV:VOLT 5 TRIG:ACQ:LEV:VOLT MAX TRIG:ACQ:LEV 2
Query Syntax	TRIGger:SEQuence2:LEVel:VOLTage? TRIGger:ACQuire:LEVel:VOLTage?
Returned Parameters	<NR3>
Related Commands	TRIG:SEQ2:LEV:CURR TRIG:SEQ2:HYST:VOLT

TRIGger:SEQuence2:SLOPe:CURRent TRIGger:ACQuire:SLOPe:CURRent

Keysight 66312A, 66332A Only

This command sets the slope of an internally triggered current measurement.

POSitive triggering occurs on the rising edge.
NEGative triggering occurs on the falling edge.
EITHer triggering occurs on either edge.

Command Syntax	TRIGger:SEQuence2:SLOPe:CURRent<slope> TRIGger:ACQuire:SLOPe:CURRent<slope>
Parameters	EITHer POSitive NEGative
*RST Value	EITHer
Examples	TRIG:SEQ2:SLOP:CURR POS TRIG:ACQ:SLOP:CURR EITH
Query Syntax	TRIGger:SEQuence2:SLOPe:CURRent? TRIGger:ACQuire:SLOPe:CURRent?
Returned Parameters	<CRD>
Related Commands	TRIG:SEQ2:SLOP:VOLT

TRIGger:SEQuence2:SLOPe:VOLTage TRIGger:ACQuire:SLOPe:VOLTage

Keysight 66312A, 66332A Only

This command sets the slope of an internally triggered voltage measurement.

POSitive triggering occurs on the rising edge.
NEGative triggering occurs on the falling edge.
EITHer triggering occurs on either edge.

Command Syntax	TRIGger:SEQuence2:SLOPe:VOLTage<slope> TRIGger:ACQuire:SLOPe:VOLTage<slope>
Parameters	EITHer POSitive NEGative
*RST Value	EITHer
Examples	TRIG:SEQ2:SLOP:VOLT POS TRIG:ACQ:SLOP:VOLT EITH
Query Syntax	TRIGger:SEQuence2:SLOPe:VOLTage? TRIGger:ACQuire:SLOPe:VOLTage?
Returned Parameters	<CRD>
Related Commands	TRIG:SEQ2:SLOP:CURR

TRIGger:SEQuence2:SOURce

TRIGger:ACQuire:SOURce

Keysight 66312A, 66332A Only

These commands select the trigger source for measurement triggers as follows:

BUS	GPIB device, *TRG, or <GET> (Group Execute Trigger)
INTernal	trigger is generated internally when the measured waveform crosses the trigger level with the selected slope.

Command Syntax	TRIGger:SEQuence2:SOURce<source> TRIGger:ACQuire:SOURce<source>
Parameters	BUS INTernal
*RST Value	INTernal
Examples	TRIG:ACQ:SOUR BUS
Query Syntax	TRIGger:SEQuence2:SOURce? TRIGger:ACQuire:SOURce?
Returned Parameters	<CRD>

TRIGger:SEQuence1:DEFine

TRIGger:SEQuence2:DEFine

TRIGger:SEQuence2:DEFine applies to Keysight 66312A, 66332A Only

These commands define the names that are aliased to trigger sequences 1 and 2. The command accepts only ACQuire for sequence 2 and TRANSient for sequence 1 as predefined names. The query allows the user to query the instrument names aliased to sequences 1 and 2.

Command Syntax	TRIGger:SEQuence1:DEFine TRANSient TRIGger:SEQuence2:DEFine ACQuire
Parameters	TRANSient, ACQuire
Examples	SEQ1:DEF ACQ SEQ2:DEF TRAN
Query Syntax	TRIGger:SEQuence1:DEFine? TRIGger:SEQuence2:DEFine?
Returned Parameters	<CRD>
Related Commands	TRIG:SEQ2:ACQ TRIG:SEQ1:TRAN

*TRG

This common command generates a trigger when the trigger subsystem has BUS selected as its source. The command has the same affect as the Group Execute Trigger (<GET>) command.

In RS-232 mode, this command emulates some of the functionality of the IEEE-488 Group Execute Trigger command.

Command Syntax	*TRG
Parameters	None
Related Commands	ABOR INIT TRIG[:IMM] <GET>

SCPI Conformance Information

SCPI Version

The Keysight Dynamic Measurement DC Source conforms to SCPI Version 1995.0.

SCPI Confirmed Commands

ABOR	SENS:SWE:POIN
CAL:DATA	SENS:SWE:TINT
CAL:STAT	STAT:OPER[:EVEN]?
DISP[:WIND][:STAT]	STAT:OPER:COND?
DISP[:WIND]:TEXT[:DATA]	STAT:OPER:ENAB
INIT[:IMM]:SEQ NAME	STAT:OPER:NTR
INIT:CONT:SEQ NAME	STAT:OPER:PTR
MEAS FETC:ARR:CURR[:DC]?	STAT:PRES
MEAS FETC:ARR:VOLT[:DC]?	STAT:QUES[:EVEN]?
MEAS FETC[:SCAL]:CURR[:DC]?	STAT:QUES:COND?
MEAS FETC[:SCAL]:CURR:HIG?	STAT:QUES:ENAB
MEAS FETC[:SCAL]:CURR:LOW?	STAT:QUES:NTR
MEAS FETC[:SCAL]:CURR:MAX?	STAT:QUES:PTR
MEAS FETC[:SCAL]:CURR:MIN?	SYST:ERR?
MEAS FETC[:SCAL]:VOLT[:DC]?	SYST:LANG
MEAS FETC[:SCAL]:VOLT:HIG?	SYST:VERS?
MEAS FETC[:SCAL]:VOLT:LOW?	TRIG[:SEQ1 :TRAN][:IMM]
MEAS FETC[:SCAL]:VOLT:MAX?	TRIG[:SEQ1 :TRAN]:SOUR
MEAS FETC[:SCAL]:VOLT:MIN?	TRIG:SEQ2 ACQ[:IMM]
OUTP[:STAT]	TRIG:SEQ2 ACQ:SOUR
OUTP:PROT:CLE	TRIG:SEQ:DEF
OUTP:PROT:DEL	*CLS
[SOUR]:CURR[:LEV][:IMM][:AMPL]	*ESE*ESE?*ESR?
[SOUR]:CURR[:LEV]:TRIG[:AMPL]	*IDN?
[SOUR]:CURR:PROT:STAT	*OPC*OPC?*OPT?
[SOUR]:VOLT[:LEV][:IMM][:AMPL]	*PSC*PSC?
[SOUR]:VOLT[:LEV]:TRIG[:AMPL]	*RCL*RST
[SOUR]:VOLT:PROT	*SAV*SRE*STB?
SENS:CURR[:DC]:RANG[:UPP]	*TRG*TST?
SENS:FUNC	*WAI
SENS:SWE:OFFS:POIN	

Non-SCPI Commands

CAL:CURR[:SOUR][:DC][:POS]	OUTP:DFI:SOUR
CAL:CURR[:SOUR][:DC]:NEG	OUTP:PON:STAT
CAL:MEAS[:DC]:LOWR	OUTP:REL[:STAT]
CAL:MEAS:AC	OUTP:REL:POL
CAL:LEV	OUTP:RI:MODE
CAL:PASS	SENS:CURR:DET
CAL:SAVE	[SOUR]:DIG:DATA[:VAL]
CAL:VOLT[:DC]	[SOUR]:DIG:FUNC
CAL:VOLT:PROT	TRIG:SEQ2 ACQ:COUN:CURR :VOLT
DISP[:WIND]:MODE	TRIG:SEQ2 ACQ:HYST:CURR :VOLT
MEAS FETC[:SCAL]:CURR:ACDC?	TRIG:SEQ2 ACQ:LEV:CURR :VOLT
MEAS FETC[:SCAL]:VOLT:ACDC?	TRIG:SEQ2 ACQ:SLOP:CURR :VOLT
OUTP:DFI[:STAT]	

Compatibility Language

Introduction

The Keysight power supplies covered by this manual are programmatically compatible with the HP/Keysight 6632A, 6633A, and 6634A dc power supplies. This means that by using COMPatibility language mode you can program these newer dc sources over the GPIB using COMPatibility commands.

To switch from SCPI commands to COMPatibility commands (and vice versa), use the SYST:LANG command, as documented in chapter 4. The language setting is saved in non-volatile memory.

Table B-2 summarizes the COMPatibility commands that program the supplies. You may need to refer to the HP/Keysight Series 6632, 6633A, and 6634A Operating Guide (p/n 5957-6360) for complete information on the COMPatibility commands.

The rest of this appendix discusses the COMPatibility language status system, and the COMPatibility language error codes.

Note: For complete information on the Compatibility programming language, order the HP/Keysight 6632A/6633A/6634A Operating manual, p/n 5957-6360.

Table B-1. COMPatibility Power-on Settings

Command	Setting	Command	Setting
DC	1 (ON)	POL	1 (normal)
DLY	8 ms (fast) 80 ms (normal)	PON	last stored value
DSP	1 (ON)	RELAY	1 (close)
ISSET	0.04 A (6631B) 0.02 A (6632B) 0.008 A (6633B) 0.004 A (6634B)	RLYPON	1 (close)
OCP	OFF	SRQ	0
OUT	1 (ON)	UNMASK	0
OVSET	MAX	VSET	0 V

Table B-2. COMPatibility Commands

Compatibility Command	Description	Similar SCPI Command
ASTS?	This command reads the contents of the accumulated status register, which stores any bit condition entered in the status register since the accumulated status register was last read, regardless of whether the condition still exists. Data Representation: ZZZZD	STAT:OPER? STAT:QUES? *ESE?
CLR	This command initializes the dc source to the power-on state. It also resets the PON bit in the serial poll register. The command performs the same function as the Device Clear (DCL) interface message.	*RST
DC 0 1	Only applies to units with Relay Option 760. This command enables or disables the output without affecting the state of the output relays. Initial condition: DC 1	OUTP:STAT[:NOR] 0 1 OFF ON
DLY <n>	This command programs the delay time between the programming of an output change that produces a CV, CC, or an UNREG condition, and the recording of that condition by the status registers. This can be used to prevent false triggering of the OverCurrent Protection feature (OCP). Initial delay: 0.08s (Normal); 0.008s (Fast).	OUTP:PROT:DEL
DSP 0 1	This command enables or disables the dc source's front panel display. Initial condition: DSP 1	DISP 0 1 OFF ON
ERR?	This command determines the type of programming error detected by the dc source. A remote programming error sets the ERR bit in the status register, which can be enabled by UNMASK to request service.	SYST:ERR?
FAULT?	This command reads which bits have been set in the fault register. A bit is set in the fault register when the corresponding bit in the status register changes from inactive to active AND the corresponding bit in the mask register has been enabled. The fault register is reset only after it has been read. The decimal equivalent of the total bit weight of all enabled bits is returned. Data Representation: ZZZZD	STAT:OPER? STAT:QUES? *ESE
ID?	This command causes the dc source to report its model number and any options that affect the dc source's output. Data Representation: Keysight663XA	*IDN?
IOUT?	This command measures and returns the actual output current. Data Representation: SD.DDDD	MEAS:CURRE?
ISSET <n>	This command programs the output current. See Table 4-3 for the programming range of this command. Initial condition: see Table B-1	CURR
OCP 0 1	This command enables or disables the dc source's overcurrent protection. If this function is enabled and the dc source goes into CC mode, the output of the dc source is disabled. Initial condition: OCP 0	CURR:PROT:STAT 0 1 OFF ON
OUT 0 1	This command enables or disables the dc source's output. The dc source will be able to implement commands even while the output is disabled. Initial condition: OUT 1	OUTP:STAT 0 1 OFF ON

Table B-2. COMPatibility Commands (continued)

Compatibility Command	Description	Similar SCPI Command
OVSET <n>	This command programs the overvoltage protection. See Table 4-3 for the programming range of this command. Initial condition: MAX	VOLT:PROT
POL 0 1	Only applies to units with Option 760. This command sets the polarity of the output relays to either normal (1) or inverted (0). Initial condition: POL 1	OUTP:REL:POL 0 1
PON 0 1	This command enables (1) or disables (0) SRQ at power-on. Initial condition: last programmed value	PSC 0 1
RELAY 0 1	Only applies to units with Relay Option 760. This command opens (0) or closes (1) the output relays without affecting the programmed output state of the unit. Initial condition: RELAY 1	OUTP:REL 0 1
RLYPON 0 1	Only applies to units with Relay Option 760. This command opens (0) or closes (1) the output relays at power-on without affecting the programmed output state of the unit. Initial condition: RLYPON 1	RCL 0
ROM?	This command returns the ROM version of the dc source. Data Representation: AAA AAA	*IDN?
RST	This command resets the dc source if the output is disabled by the output protection circuits.	OUTP:PROT:CLE
SENS:CURRE:RANG <n>	This command sets the current measurement range of the dc source. See Table 4-3 for the programming range of this command. Initial condition: MAX	SENS:CURRE:RANG
SENS:SWE:POIN <n>	This command defines the number of data points in a measurement sweep. Initial condition: 32	SENS:SWE:POIN
SENS:SWE:TINT <n>	This command defines the time period between measurement samples. Initial condition: 15.6 μ s.	SENS:SWE:TINT
SRQ 0 1	These commands enable or disable the dc source's ability to request service from the controller for fault conditions. UNMASK defines which conditions are defined as faults. Initial condition: SRQ 0	*SRE
STS?	This command reads the contents of the status register, which maintains the present status of the dc source. Data Representation: ZZZZD	STAT:OPER:COND? STAT:QUES:COND? *ESE?
SYST:LANG	This command causes the alternate language to become active and to be stored in nonvolatile memory. In this case, the commands are equivalent. After being shut off, the dc source will resume in the last-selected language when power is restored. The parameter must be either COMP or SCPI.	SYST:LANG
TEST?	This command causes the dc source to run selftest and report any detected failures. Data Representation: ZZZZD	*TST?

Table B-2. COMPatibility Commands (continued)

Compatibility Command	Description	Similar SCPI Command
UNMASK xxx	These commands determine the conditions that will set bits in the fault register, allowing the operator to define the conditions that will be reported as fault. Fault conditions can be enabled by sending the decimal equivalent of the total bit weight of all conditions to be enabled.	STAT:OPER:ENAB STAT:QUES:ENAB *ESE
VOUT?	This command measures and returns the actual output voltage. Data Representation: SZZD.DD; (SZD.DDD for 6634B only)	MEAS:VOLT?
VSET <n>	This command programs the output voltage. See Table 4-3 for the programming range of this command. Initial condition: 0 V	VOLT

A = Alpha

D = Digit

S = Sign (blank for positive and – for negative)

Z = Digit with leading zeros output as spaces

Table B-3. COMPatibility Errors

Error	NumberError String [Description/Explanation/Examples]
ERR 0	No error
ERR 1	EEPROM save failed [Data write to non-volatile memory failed]
ERR 2	Second PON after power-on [More than one PON command received after power-on. Only one is allowed.]
ERR 4	RLYPON sent with no relay option present [A RLYPON command was sent with no relay option present.]
ERR 5	No relay option present [A relay option command was sent with no relay option present.]
ERR 8	Addressed to talk and nothing to say [The unit was addressed to talk without first receiving a query.]
ERR 10	Header expected [A non-alpha character was received when a header was expected.]
ERR 11	Unrecognized header [The string of alpha characters received was not a valid command.]
ERR 20	Number expected [A non-numeric character was received when a number was expected.]
ERR 21	Number Syntax [The numeric character received did not represent a proper number.]
ERR 22	Number out of internal range [The number received was too large or small to be represented in internal format.]
ERR 30	Comma [A comma was not received where one was expected.]
ERR 31	Terminator expected [A valid terminator was not received where one was expected.]
ERR 41	Parameter Out [The number received exceeded the limits for its associated command.]
ERR 42	Voltage Programming Error [The programmed value exceeded the valid voltage limits.]
ERR 43	Current Programming Error [The programmed value exceeded the valid current limits.]
ERR 44	Overvoltage Programming Error [The programmed value exceeded the valid overvoltage limits.]
ERR 45	Delay Programming Error [The programmed value exceeded the valid delay limits.]
ERR 46	Mask Programming Error [The programmed value exceeded the fault mask limits.]
ERR 51	EEPROM Checksum [EEPROM failed, or a new uncalibrated EEPROM was installed.]

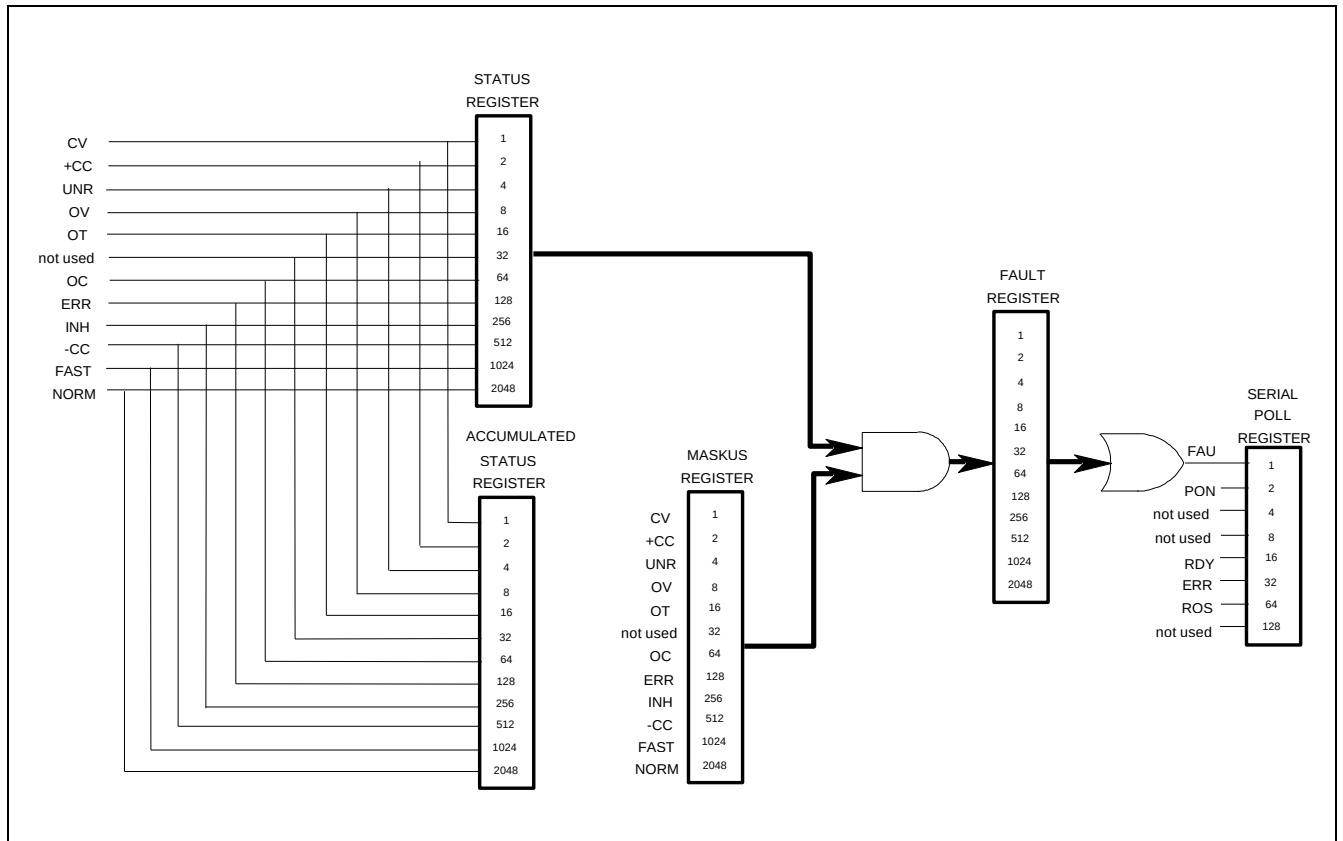


Figure B-1. COMPatibility Status Model

Table B-4. Bit Assignment of Status, Astatus, Fault, & Mask Registers

Bit Position	11	10	9	8	7	6	5	4	3	2	1	0
Bit Name	NORM	FAST	-CC	INH	ERR	OC	not used	OT	OV	UNR	+CC	CV
Bit Weight	2048	1024	512	256	128	64	32	16	8	4	2	1

CV = The unit is operating in constant voltage mode.

CC+ = The unit is operating in constant current mode.

UNR = The output of the unit is unregulated.

OV = The overvoltage protection circuit has tripped.

OT = The over-temperature protection circuit has tripped.

OC = The overcurrent protection circuit has tripped.

ERR = A programming error has occurred. Use ERR? to clear.

CC = The unit is operating in negative constant current mode.

INH = The external remote inhibit signal has turned the output off.

FAST = The output is in Fast operating mode. (Keysight 66332A, 6631B, 6632B, 6633B, 6634B only)

NORM = The output is in Normal operating mode. (Keysight 66332A, 6631B, 6632B, 6633B, 6634B only)

Table B-5. Bit Configuration of Serial Poll Register

Bit Position	7	6	5	4	3	2	1	0
Bit Name	not used	RQS	ERR	RDY	not used	not used	PON	FAU
Bit Weight		64	32	16			2	1

RQS = The dc source has generated a service request. Use a serial poll to clear.

ERR = Same as ERR bit in Status register. Use ERR? to clear.

RDY = This bit cleared when unit busy processing commands. Set when processing complete.

PON = A Power-on has occurred. Use CLR to clear.

FAU = A bit has been set in the Fault register. Use FAULT? to clear.

Error Messages

Error Number List

This appendix gives the error numbers and descriptions that are returned by the dc source. Error numbers are returned in two ways:

- ◆ Error numbers are displayed on the front panel
- ◆ Error numbers and messages are read back with the SYSTem:ERRor? query. SYSTem:ERRor? returns the error number into a variable and returns two parameters: an NR1 and a string.

The following table lists the errors that are associated with SCPI syntax errors and interface problems. It also lists the device dependent errors. Information inside the brackets is not part of the standard error message, but is included for clarification.

When errors occur, the Standard Event Status register records them as follows:

Bit Set	Error Code	Error Type	Bit Set	Error Code	Error Type
5	-100 thru -199	Command	3	-300 thru -399 or 1 thru 32767	Device-dependent
4	200 thru -299	Execution	2	-400 thru -499	Query

Table C-1. Error Numbers

Error Number	Error String [Description/Explanation/Examples]
-100	Command error [generic]
-101	Invalid character
-102	Syntax error [unrecognized command or data type]
-103	Invalid separator
-104	Data type error [e.g., "numeric or string expected, got block data"]
-105	GET not allowed
-108	Parameter not allowed [too many parameters]
-109	Missing parameter [too few parameters]
-112	Program mnemonic too long [maximum 12 characters]
-113	Undefined header [operation not allowed for this device]
-121	Invalid character in number [includes "9" in octal data, etc.]
-123	Numeric overflow [exponent too large; exponent magnitude >32 k]
-124	Too many digits [number too long; more than 255 digits received]
-128	Numeric data not allowed
-131	Invalid suffix [unrecognized units, or units not appropriate]
-138	Suffix not allowed

Table C-1. Error Numbers (continued)

Error Number	Error String [Description/Explanation/Examples]
-141	Invalid character data [bad character, or unrecognized]
-144	Character data too long
-148	Character data not allowed
-150	String data error
-151	Invalid string data [e.g., END received before close quote]
-158	String data not allowed
-160	Block data error
-161	Invalid block data [e.g., END received before length satisfied]
-168	Block data not allowed
-170	Expression error
-171	Invalid expression
-178	Expression data not allowed
-200	Execution error [generic]
-222	Data out of range [e.g., too large for this device]
-223	Too much data [out of memory; block, string, or expression too long]
-224	Illegal parameter value [device-specific]
-225	Out of memory
-270	Macro error
-272	Macro execution error
-273	Illegal macro label
-276	Macro recursion error
-277	Macro redefinition not allowed
-310	System error
-350	Too many errors [errors beyond 9 lost due to queue overflow]
-400	Query error [generic]
-410	Query INTERRUPTED [query followed by DAB or GET before response complete]
-420	Query UNTERMINATED [addressed to talk, incomplete programming message received]
-430	Query DEADLOCKED [too many queries in command string]
-440	Query UNTERMINATED [after indefinite response]
0	No error
1	Non-volatile RAM RD0 section checksum failed
2	Non-volatile RAM CONFIG section checksum failed
3	Non-volatile RAM CAL section checksum failed
4	Non-volatile RAM STATE section checksum failed
5	Non-volatile RST section checksum failed
10	RAM selftest
11	VDAC/IDAC selftest 1
12	VDAC/IDAC selftest 2
13	VDAC/IDAC selftest 3
14	VDAC/IDAC selftest 4
15	OVDAC selftest
80	Digital I/O selftest error

Table C-1. Error Numbers (continued)

Error Number	Error String [Description/Explanation/Examples]
213	Ingrd receiver buffer overrun
216	RS-232 receiver framing error
217	RS-232 receiver parity error
218	RS-232 receiver overrun error
220	Front panel uart overrun
221	Front panel uart framing
222	Front panel uart parity
223	Front panel buffer overrun
224	Front panel timeout
401	CAL switch prevents calibration
402	CAL password is incorrect
403	CAL not enabled
404	Computed readback cal constants are incorrect
405	Computed programming cal constants are incorrect
406	Incorrect sequence of calibration commands
407	CV or CC status is incorrect for this command
408	Output mode switch must be in NORMAL position
601	Too many sweep points
602	Command only applies to RS-232 interface
603	CURRent or VOLTage fetch incompatible with last acquisition
604	Measurement overrange

Example Programs

Introduction

The example programs in this section are intended to show how some of the same dc source functions can be programmed to each of the following GPIB interfaces:

1. HP Vectra PC controller with Keysight 82335A GPIB Interface Command Library
2. IBM PC controller with National Instruments GPIB-PCII Interface/Handler
3. Keysight controller with BASIC Language System

Assigning the GPIB Address in Programs

The dc source address cannot be set remotely. It must be set using the front panel Address key. Once the address is set, you can assign it inside programs. The following examples assume that the GPIB select code is 7, and the dc source is assigned to the variable PS.

```
1070 PS=706                !Keysight 82335A Interface
1070 ASSIGN @PS TO 706      !BASIC Interface
```

For systems using the National Instruments DOS driver, the address is specified in the software configuration program (IBCONFIG.EXE) and assigned a symbolic name. The address then is referenced only by this name within the application program (see the National Instruments GPIB documentation).

Types of DOS Drivers

The Keysight 82335A and National Instruments GPIB are two popular DOS drivers. Each is briefly described here. See the software documentation supplied with the driver for more details.

Keysight 82335A Driver

For GW-BASIC programming, the GPIB library is implemented as a series of subroutine calls. To access these subroutines, your application program must include the header file SETUP.BAS, which is part of the DOS driver software.

SETUP.BAS starts at program line 5 and can run up to line 999. Your application programs must begin at line 1000. SETUP.BAS has built-in error checking routines that provide a method to check for GPIB errors during program execution. You can use the error-trapping code in these routines or write your own code using the same variables as used by SETUP.BAS.

National Instruments GPIB Driver

Your program must include the National Instruments header file DECL.BAS. This contains the initialization code for the interface. Prior to running any applications programs, you must set up the interface with the configuration program (IBCONF.EXE).

D - Example Programs

Your application program will not include the dc source's symbolic name and GPIB address. These must be specified during configuration (when you run IBCONF.EXE). Note that the primary address range is from 0 to 30 but any secondary address must be specified in the address range of 96 to 126. The dc source expects a message termination on EOI or line feed, so set EOI w/last byte of Write. It is also recommended that you set Disable Auto Serial Polling.

All function calls return the status word IBSTA%, which contains a bit (ERR) that is set if the call results in an error. When ERR is set, an appropriate code is placed in variable IBERR%. Be sure to check IBSTA% after every function call. If it is not equal to zero, branch to an error handler that reads IBERR% to extract the specific error.

Error Handling

If there is no error-handling code in your program, undetected errors can cause unpredictable results. This includes "hanging up" the controller and forcing you to reset the system. Both of the above DOS drivers have routines for detecting program execution errors. Error detection should be used after every call to a subroutine.

BASIC Controllers

The BASIC Programming Language provides access to GPIB functions at the operating system level. This makes it unnecessary to have the header files required in front of DOS applications programs. Also, you do not have to be concerned about controller "hangups" as long as your program includes a timeout statement. Because the dc source can be programmed to generate SRQ on errors, your program can use an SRQ service routine for decoding detected errors. The detectable errors are listed in Appendix C.

Example 1. HP Vectra PC Controller Using Keysight 82335 Interface

```
5      '----- Merge SETUP.BAS here -----
1000  MAX.ELEMENTS=2 :ACTUAL.ELEMENTS=0 :MAX.LENGTH=80 :ACT.LENGTH=0
1005  DIM OUTPUTS(2) :CODES$=SPACE$(40)
1010  ISC=7 :PS=706
1015  '
1020  'Set up the DC Source Interface for DOS driver
1025  CALL IORESET (ISC)                'Reset the interface
1030  IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR
1035  TIMEOUT=3
1040  CALL IOTIMEOUT (ISC, TIMEOUT)      'Set timeout to 3 seconds
1045  IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR
1050  CALL IOCLEAR (ISC)                'Clear the interface
1055  IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR
1060  CALL IOREMOTE (ISC)               'Set dc source to remote
mode
1065  IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR
1070  '
1075  'Program dc source to CV mode with following voltage and current
1080  CODES$ = "VOLTAGE MAX;CURRENT MAX" :GOSUB 2000
```

```

1085 '
1090 'Query dc source outputs CURRENT?" :GOSUB 2000 :GOSUB 3000
1100 VOUT = OUTPUTS(1)
1105 IOUT = OUTPUTS(2)
1110 PRINT "The output levels are "VOUT" Volts and "IOUT" Amps"
1115 '
1120 'Program triggered current level to value insufficient to maintain
1125 'supply within its CV operating characteristic
1130 CODES$ = "CURR:TRIG MIN" :GOSUB 2000
1135 '
1140 'Set operation status mask to detect mode change from CV to CC
1145 CODES$ = "STAT:OPER:ENAB 1024;PTR 1024" :GOSUB 2000
1150 '
1155 'Enable Status Byte OPER summary bit
1160 CODES$ = "*SRE 128" :GOSUB 2000
1165 '
1170 'Arm trigger circuit and send trigger to dc source
1175 CODES$ = "INITIATE:SEQUENCE1;TRIGGER" :GOSUB 2000
1180 '
1185 'Wait for supply to respond to trigger
1190 FOR I= 1 to 100 :NEXT I
1195 '
1200 'Poll for interrupt caused by change to CC mode and print to
screen
1205 CALL IOS POLL (PS,RESPONSE)
1210 IF (RESPONSE AND 128) <> 128 THEN GOTO 1240 'No OPER event to
report
1215 CODES$ = "STATUS:OPER:EVEN?" :GOSUB 2000 'Query status oper
register
1220 CALL IOENTER (PS,OEVENT) 'Read back event bit
1225 IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR
1230 IF (OEVENT AND 1024) = 1024 THEN PRINT "Supply switched to CC
mode."
1240 'Clear the status circuit
1245 CODES$ = "*CLS" :GOSUB 2000
1250 FOR I = 1 TO 100 :NEXT I 'Wait for supply to
clear
1255 '
1260 'Disable output and save present state in location 2
1265 CODES$ = "OUTPUT OFF;*SAV 2" :GOSUB 2000
1270 END
1275 '
2000 'Send command to dc source
2005 LENGTH = LEN(CODES$)
2010 CALL IOOUTPUTS (PS,CODES$,LENGTH) 'Send command to
interface
2015 IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR 'SETUP.BAS error
trap
2020 RETURN
2025 '
3000 'Get data from dc source
3005 CALL IOENTERA (PS,OUTPUTS(1),MAX.ELEMENTS,ACTUAL.ELEMENTS)
3010 IF PCIB.ERR <> NOERR THEN ERROR PCIB.BASERR
3015 RETURN

```


D - Example Programs

Example 2. IBM Controller Using National Interface

```
990  '----- Merge DECL.BAS here -----
1000 'DC Source Variable = PS% ; Stand-Alone Address = 706
1005 CODES$=SPACE$(50):MODE$=SPACE$(5):OEVENT$=SPACE$(20)
1010 D$=SPACE$(60):OUTPUT$=SPACE$(40):BDNAME$="PS%"
1015 DIM OUTPUT(2)
1020 '
1025 'Set up dc source interface for DOS driver
1030 CALL IBFIND(BDNAME$,PS%)
1035 IF PS%
1040 CALL IBCLR(PS%)
1045 '
1050 'Program dc source to CV mode with following voltage and current
1055 CODES$ = "VOLTAGE MAX;CURRENT MAX" :GOSUB 2000
1060 '
1065 'Query dc source outputs and print to screen
1070 CODES$ = "MEASURE:VOLTAGE?;CURRENT?" :GOSUB 2000 :GOSUB 3000
1075 VOUT = OUTPUT(1)
1080 IOUT = OUTPUT(2)
1085 PRINT"The programmed levels are "VOUT" Volts and "IOUT" Amps"
1090 '
1095 'Program triggered current level to value insufficient to maintain
1100 'supply within its CV operating characteristic
1105 CODES$ = "CURR:TRIG MIN" :GOSUB 2000
1110 '
1115 'Set operation status mask to detect mode change from CV to CC
1120 CODES$ = "STAT:OPER:ENAB 1024;PTR 1024" :GOSUB 2000
1125 '
1130 'Enable Status Byte OPER summary bit
1135 CODES$ = "*SRE 128" :GOSUB 2000
1140 '
1145 'Arm trigger circuit and send trigger to dc source
1150 CODES$ = "INITIATE:SEQUENCE1;TRIGGER" :GOSUB 2000
1160 'Wait for supply to respond to trigger
1165 FOR I= 1 to 100 :NEXT I
1170 '
1175 'Poll for interrupt caused by change to CC mode and print to screen
1180 SPOL%=0
1185 CALL IBRSP(PS%,SPOL%)
1190 IF (SPOL% AND 128) = 128 THEN POLL = 1 'Set interrupt flag on
OPER bit
1195 IF POLL <> 1 THEN GOTO 1230 'No interrupt to
service
1200 "CODES$ = "STAT:OPER:EVEN?" :GOSUB 2000 'Query status oper
register
1205 CALL IBRD(PS%,OEVENT$) 'Read back event bit
1210 IF IBSTA%
1215 OEVENT=VAL(OEVENT$)
1220 IF (OEVENT AND 1024) = 1024 THEN PRINT "Supply switched to CC mode."
```

Example Programs - D

```

1225 '
1230 'Clear status circuit
1235 CODES$="*CLS" :GOSUB 2000
1240 FOR I=1 TO 50 :NEXT I 'Wait for supply to clear
1245 '
1250 'Disable output and save present state to location 2
1255 CODES$ = "OUTPUT OFF;*SAV 2" :GOSUB 2000
1260 END
1265 '
2000 'Send command to dc source
2005 CALL IBWRT(PS%,CODES$)
2010 IF IBSTAT%
2015 RETURN
1250 'Disable output and save present state to location 2
1255 CODES$ = "OUTPUT OFF;*SAV 2" :GOSUB 2000
1260 END
1265 '
2000 'Send command to dc source
2005 CALL IBWRT(PS%,CODES$)
2010 IF IBSTAT%
2015 RETURN
2020 '
2100 'Error detection routine
2105 PRINT "GPIB error. IBSTAT% = HEX$(IBSTAT%)
2110 PRINT " IBERR% = ";IBERR%" in line ";ERL
2115 STOP
2120 '
3000 'Get data from dc source
3005 CALL IBRD(PS%,OUTPUT$)
3010 IF IBSTA%
3015 I=1 'Parse data string
3020 X=1
3025 C=INSTR(I,OUTPUT$,";")
3030 WHILE C <> 0
3035 D$=MID$(OUTPUT$,I,C-I)
3040 OUTPUT(X)=VAL(D$) 'Get values
3045 I=C+1
3050 C=INSTR(I,OUTPUT$,";")
3055 X=X+1
3060 WEND
3065 D$=RIGHT$(OUTPUT$,LEN(OUTPUT$)-(I-1))
3070 OUTPUT(X)=VAL(D$)
3075 OUTPUT$=SPACE$(40) 'Clear string
3080 RETURN

```

D - Example Programs

Example 3. Controller Using BASIC

```
1000 !Dc source at stand-alone address = 706
1010 OPTION BASE 1
1020 DIM Response$(80)
1030 ASSIGN @Ps TO 706
1040 CLEAR SCREEN
1050 !
1060 PRINT "Disconnect any load from output terminals and press
CONTINUE..."
1070 PAUSE
1080 !
1090 !Program dc source with following voltage level and current limit
1100 OUTPUT @Ps;"VOLT MAX"
1110 OUTPUT @Ps;"CURR MAX"
1120 OUTPUT @Ps;"OUTP ON"
1130 !
1140 !Query dc source outputs and print to screen
1150 OUTPUT @Ps;"MEAS:VOLT?;CURR?"!Query output levels
1160 ENTER @Ps;Vout,Iout
1170 PRINT "The output levels are ";Vout;" Volts and ";Iout;" Amps"
1180 !
1190 !Program current triggered level to a value insufficient to
maintain
1200 !supply within its CV operating characteristic
1210 OUTPUT @Ps;"CURR:TRIG MIN"
1220 !
1230 !Set operation status mask to detect mode change from CV to CC
1240 OUTPUT @Ps;"STAT:OPER:ENAB 1024;PTR 1024"
1250 !
1260 !Enable Status Byte OPER summary bit
1270 OUTPUT @Ps;"*SRE 128"
1280 !
1290 !Arm trigger circuit and send trigger to dc source
1300 OUTPUT @Ps;"INIT:NAME TRAN"
1310 OUTPUT @Ps;"TRIGGER"
1320 !Poll for interrupt caused by change to CC mode and print to screen
1330 PRINT "Connect 1K ohm load and hit CONTINUE..."
1340 PAUSE
1350 Response=SPOLL(@Ps)
1360 IF NOT BIT(Response,7) THEN !No OPER event to report
1370 PRINT "Supply not in CC mode!!!"
1380 GOTO 1320
1390 END IF
1400 OUTPUT @Ps;"STAT:OPER:EVEN?"!Query status operation register
1410 ENTER @Ps;Oevent!Read back event bit
1420 IF BIT(Oevent,10) THEN PRINT "Supply switched to CC mode."
1430 !
1440 !Clear status
1450 OUTPUT @Ps;"*CLS"
1460 !
1470 !Disable output and save present state in location 2
1480 OUTPUT @Ps;"OUTPUT OFF;*SAV 2"
1490 PRINT "Program terminated."
1500 END
```

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